INDIVIDUAL ASSIGNMENT PART 1

TECHNOLOGY PARK MALAYSIA

CT015-3-2

DESIGN METHODS

APD2F2409SE

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Table of Contents

1.0 Project Background

As technology advances, it makes people more efficient, easier, faster, and more organized in the context of project management. It helps users plan, track, and manage their tasks, schedules, deadlines, and team collaboration. Notion is primarily a planner management application and cloud-based productivity and collaboration tool. It's designed to be a single space for capturing thoughts, managing projects, and organizing information, whether for personal or professional use (Lau, 2024). In 2016, Notion was released as the first generation of desktop on San Francisco, and the android app released in 2018. Notion is one of the best planner systems in the world. It provides many useful functions to let the users easily to mange their project and easily to share and access with the different area users to complete the task with good collaboration and communication on the teamwork. Furthermore, it also can save time using useful functional features to create content, take notes, organize projects, automate workflows, or assign task (Deer & Nevena Radulović, 2023). According to Ivan Zhao, Notion has passed 100 million users in September 2023. It is more than its competitor, Trello.

However, despite its comprehensive functionality, Notion faces challenges, particularly when handling complex projects. After users entry their tasks and schedules, they are not easy to clearly understand tasks and projects progress when dealing with complex pages and multiple nested documents. Raymond was said in 2023, this inefficiency can significantly impact productivity, especially for users managing multiple projects or collaborating with large teams. While using the visualize graph in project dashboards can ensure the user clearly visualizes overall tasks progress and let users make decisions efficiently with easy management. By using the Application Programming Interface (API) to connect the frontend graph components and backend database in real time analysis data. This also can improve the system without user manual updates or data entry.

Moreover, Notion lacks a dedicated and well-integrated time tracking feature compared to Trello. Trello is offering third-party time tracking plugins and power-ups that seamlessly record task durations and deadlines, Notion still relies heavily on manual input or external integrations for time management. This shortcoming limits its usefulness for teams that require precise time allocation, performance tracking, or automatic deadline reminders as part of their project planning workflow. By the external system and application to achieve the real time tracker is Clockify. Clockify is a web-based time tracker and timesheet app that lets you track work hours across projects. Notion can use the API to connect the Clockify data and real time tracker system's data to database and show the result of each task and projects. The time tracker also needs to link the hardware, software, and system which the users open or start their work. This can let the users to improve their time management and productivity for eliminating time wasting activities and accurate project estimation for setting the projects and tasks dateline

(Sorokina et al., 2021).

Hence, this project aims to improve the planner management system which is Notion by developing visualize graph functions in projects overall dashboard and the time tracking function to provide more efficiency, easy management fast navigation and efficient time tracking in Notion Application.

2.0 Software Development Methods

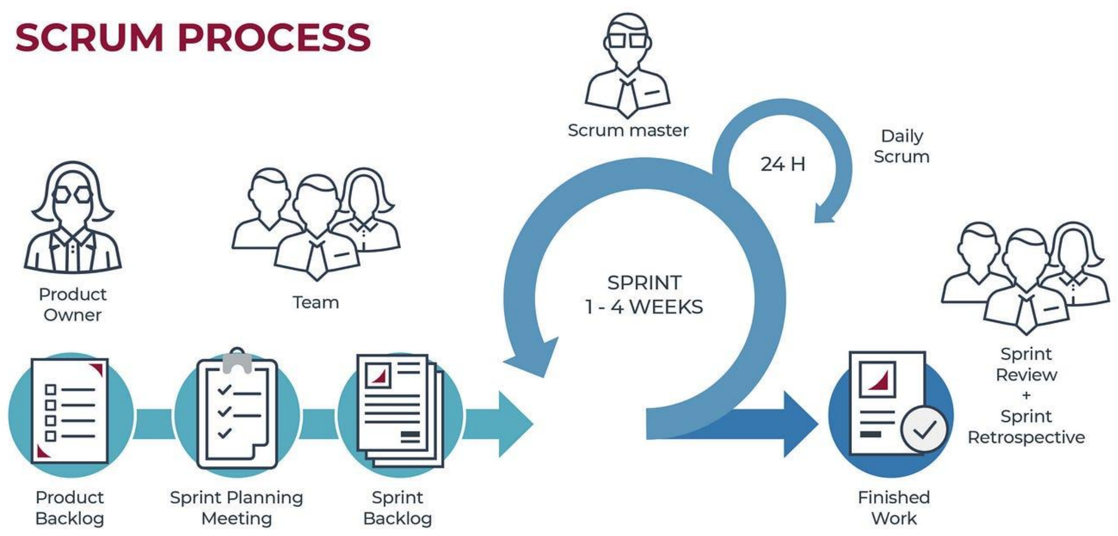


Figure 1 Scrum Phases Diagram

A software development methodology is a process by which developers design, implement and test new computer programs. Following a methodology benefits developers because these processes lay out a structured sequence of steps that guide professionals through each stage of development. It have many methods to consider the project such as Agile, Scrum, Waterfall, and Spiral. In these methodologies, Scrum Methodology is the suitable methodologies to develop the Notion application. Scrum is a lightweight Agile framework that focuses on delivering functional software through iterative and incremental development cycles called "Sprints." Each Sprint typically for one to four weeks and results in a potentially shippable product increment, allowing teams to deliver value quickly and adapt to evolving requirements (Artem Gurnov, 2022). It emphasizes continuous feedback, collaboration, and transparency, making it highly suitable for this project, which involves developing a visualization graph, an overall dashboard, and an integrated time tracking function. Each Sprint will also include planning the development of these features, enabling the team to regularly collect user feedback and iterate on features based on real user needs, usability testing, etc. Moreover, Scrum promotes the roles of the benefited owner, scrum master, and development team thus facilitating clarity about roles and the consequent efficient working together. This structure will help reduce user feedback loops, catch detections at an early stage, and improve the front end and back end in accordance with actual use cases. Scrum's responsiveness and focus on the user have made it the perfect technique to ensure the final Notion enhancement is as solid yet easy to use as possible.

3.0 Use Case Diagaram

Figure 2 Planner Management Application Notion's Use Case Diagram

4.0 Use Case Specifications

1. UC1_Create Project based with Templates or Tools

Use Case Name	Create project based with templates or tools
Actor	Planner
Description	Planner can create a new project dashboard
Precondition	• Planner must sign in to the application
Postcondition	• Planner successfully created a new project dashboard
Standard Process	1. Planner clicks 'Create New Page' button. 2. System displays an empty dashboard. 3. Planner clicks the "Template" button on side bar 4. System loading templates list 5. System displays templates list 6. Planner selects the template 7. System loads the selected project templates information 8. System displays the selected project templates information 9. Planner clicks the "Add" button to create a new project 10. System downloads the template 11. Planner clicks the project's selector bar on side bar 12. System displays project dashboard 13. Planner view projects dashboard
Alternative Path	**3a Select Personalization Tools** 3a.1. Planner selects the personalization tools. 3a.2. System download and saved selected tools. 3a.3. System displays the tools. 3a.4. System continues in step 12. **3b. Not Select Any Templates and Tools** 3b.1 Planner view the empty dashboard.
Exception Flow	**5a. Invalid Access Template** 5a.1 System failed to access template. 5a.2 System display "We couldn't find that page" message. **7a. Failed Download Template** 7a.1 System cannot download template. 7a.2 System display "Please go online to copy this block" message. 7a.3 Planner clicks "OK" button.

2. UC2_Select Workspace to View Project on Sidebar

Use Case Name	Select Project Dashboard to View Project on sidebar
Actor	Planner
Description	For planner select the project's workspace which has already been created to view the project dashboard
Precondition	• Planner must sign in to the application • Planner must create a project dashboard
Postcondition	• Planner can view the project
Standard Process	1. System displays a home page and side bar. 2. Planner selected the specific workspace on side bar 3. System Display the project dashboard and tasks 4. Planner view the project dashboard and tasks
Alternative Path	-
Exception Flow	**<u>2a. No Existing Project</u>** 2a.1 Planner need to create a project

3. UC3_Manage Task

Use Case Name	Task Manage
Actor	Planner
Description	For Planner to create, modify, or delete their task in a project dashboard.
Precondition	• Planner must create a project dashboard
Postcondition	• Planner can view the latest updated tasks in project dashboard
Standard Process	1. System displays the project dashboard and task. 2. Planner views the project dashboard and tasks. 3. Planner clicks the "New Task" button to add a task. 4. System displays the task information form. 5. Planner completed to fill the form. 6. Planner saved the task in project. 7. System displays the task title and status on project dashboard. 8. Planner views the dashboard and updated task
Alternative Path	**<u>3a. Modify Tasks Information</u>** 3a.1. Planner selects the task to modify the information. 3a.2. System displays task information forms. 3a.3. Planner continues in step 5. **<u>3b. Delete Task</u>**

	3b.1. Planner clicks more action button. 3b.2. Planner selects delete section to remove the task. 3b.3. System removes the task and moves it to trash. **<u>3c. Have Not Any Motion</u>** 3c.1 Planner have not added, modified, and deleted task.
Exception Flow	-

4. UC4_Set Time & Date

Use Case Name	Set Time & Date
Actor	Planner
Description	For planner to set the due date and time in the projects and tasks to release the timeline
Precondition	• Planner must sign in to the application • Planner must create a project dashboard • Planner must create a new task or modify the tasks.
Postcondition	• Planner set or change the date and time in the project and time
Standard Process	1. System displays the task information form 2. System retrieves and checks today's date 3. System pre-filled with today's date in date selector 4. System detects time selector 5. System displays the calendar if the time selector is off 6. Planner selects desired due date 7. System records and saved selected due date 8. System updated due date on associated projects and tasks 9. Planner view the updated
Alternative Path	**<u>4a. Time Selector Opened</u>** 4a.1. Planner opens time selector 4a.2. System displays the calendar, time selector with initial time, time zone, and time format 4a.3. Planner selects the due date, due time, time zone, and time format 4a.4. System record and saved selected due date, due time, time zone, and time format. 4a.5. System updated due date and time on associated projects and tasks 4a.6. System resumes in Step 9
Exception Flow	**7a. Invalid Date and Planner Cancel Updated Date** 7a.1. Planner inputs an invalid date 7a.2. System displays an "Invalid Date" message 7a.3. Planner cancels the date update

	7a.4. System reverts and pre-fills the previously updated date in the calendar **7b. Invalid Date and Planner Re-Enter Valid Date** 7b.1. Planner inputs an invalid date 7b.2. System displays “Invalid Date” message 7b.3. Planner enters a valid date 7b.4. System resumes at Step 7 **7c. Invalid Time and Planner Cancel Updated Time** 7c.1. Planner inputs an invalid time 7c.2. System display “Invalid Time” message’ 7c.3. Planner cancels the time update 7c.4. System reverts and pre-fills the previously updated time in the calendar **7d. Invalid Time and Planner Re-Enter Valid Time** 7d.1. Planner inputs an invalid time 7d.2. System displays “Invalid Time” message 7d.3. Planner enters a valid time 7d.4. System resumes at Step 7

5. UC5_Open Relevant File with URL Link

Use Case Name	Add relevant file with URL link
Actor	Planner
Description	For planner upload their relevant Google File, Figma, and GitHub link in specific task to access and find the file quickly.
Precondition	• Planner must sign in to the application • Planner must create a project • Planner must create a task.
Postcondition	• Planner view associated file and websites with URL link in the task information form
Standard Process	1. System displays the task. 2. Planner presses embed google drive button. 3. System displays the embed link selector bar. 4. Planner views link selector bar. 5. Planner inputs Google file link. 6. Planner clicks link button. 7. System records and saved the file link. 8. System displays saved link in link selector bar . 9. System verified link successfully. 10. System accesses Google Drive File.

	11. Planner click link selector bar within Drive Link. 12. System opens the browser to access the file.
Alternative path	-
Exception Flow	**9a: System Failed Verified Link** 9a.1. System display "Unrecognized URL" on link selector bar

6. UC6_Personal Notification & Reminder

Use Case Name	Personal notification & reminder
Actor	Planner
Description	For notifying the planner of comments, mentions, and replies by other planners
Precondition	Planner must sign in to the application
Postcondition	Planner receives notifications for relevant activities
Standard Process	1. System monitors planner-related activities. 2. System detects relevant notification trigger in new comments, mentions, and replies. 3. System checks user id available. 4. System generates notification content. 5. System display notification. 6. Planner receives notification.
Alternative path	-
Exception Flow	**3a. Planner ID Unavailable** 3a.1. System detects errors. 3a.2. System generates "User Not Found" and suggestion messages 3a.3. System continue in step 6

7. UC7_Solve Questions Using Notion AI

Use Case Name	Solve Questions Using Notion AI
Actor	Planner
Description	For planner to solve their questions by using the AI
Precondition	• Planner must sign in system
Postcondition	• Planner have received the AI's generated answers
Standard Process	1. System displays AI chatbot, text area, and submits button 2. Planner views the AI chatbot, text area, and submits button 3. Planner input the question 4. System received question 5. Notion AI model process question 6. Notion AI model generated answers 7. System displays the Notion AI model's generated answer to the planner in chatbot.

	8. Planner view the answers
Alternative Path	-
Exception Flow	**<u>5a. Planner Inputs Empty Question.</u>** 5a.1. System displays error messages 5a.2. System prompts planner to input valid questions 5a.3. Planner continues in Step 3. **<u>6a. Notion AI Retrying When Fails to Process Question</u>** 6a.1. Notion AI retrying process. 6a.2. Notion AI continues in Step 5. **<u>6b. System End When Fails to Process Question</u>** 6b.1. Notion AI generates error and invalid message. 6b.1. System resumes on Step 8

8. UC8_Searching Project and Task

Use Case Name	Searching Project and Task
Actor	Planner
Description	For planner to search associated projects in database
Precondition	• Planner must create the project. • Planner must sign in to application.
Postcondition	• Planner can view the associated projects and tasks in a list
Standard Process	1. Planner clicks the search bar button on side taskbar. 2. System displays searching selector bar and all project in a list. 3. Planner inputs the name of projects or tasks in the selector bar. 4. System compares names of projects and tasks based on alphabets. 5. System displays the projects and tasks which successfully compare. 6. Planner views list of projects and tasks.
Alternative Path	-
Exception Flow	**<u>5a. Failed Matching</u>** 5a.1 System display Not Result and AI action button **<u>6a. Real-Time Search Looping When Planner Continue Typing</u>** 6a.1 System resumes to Step 4

9. UC9_Sharing Project Dashboard

Use Case Name	Sharing Project Dashboard
Actor	Planner
Description	For planner to share project dashboard and its contents with other team members or stakeholders
Precondition	• Planner must sign in to the application • Planner must create a project dashboard
Postcondition	• Planner successfully shared with other planners
Standard Process	1. Planner selected a project 2. Planner clicks the "Share" button 3. System displays sharing dialog with options 4. Planner inputs recipient email addresses 5. Planner sets appropriate access permissions in view or edit 6. Planner clicks "Share" to confirm 7. System validates sharing permissions 8. System sends access notifications to recipients 9. System displays “Success Send Invite” message 10. System records and saved the other planner email 11. System displays other planner email in dialog 12. Other planner receives access notification 13. Other planner can open the dashboard
Alternative Path	**<u>4a. Copy Link to Share Project Dashboard</u>** 4a.1 Planner wants use link to share the project dashboards. 4a.2 Planner clicks copy button. 4a.3 System generated URL link 4a.2 System display “Copied link to clipboard” message. 4a.3 Planner sets appropriate access permissions in view or edit 4a.4. Planner send to other planner 4a.5. Other planner receive the link 4a.6. Other planner click the link 4a.7. Other planner continue in step 13
Exception Flow	**<u>5a. Invalid Email</u>** 5a.1. Planner input invalid email 5a.2. System displays invalid email

10. UC10_View Trash History

Use Case Name	View Trash History
Actor	Planner
Description	For planner to view and manage items that have been moved to trash.
Precondition	• Planner must sign in to the application.

	• Planner must create projects or task.
Postcondition	• Planner can view the deleted projects and tasks in trash history.
Standard Process	1. Planner selects trash selector bar 2. System loading data 3. System displays trash history 4. Planner view the trash history
Alternative Path	**4a. Filter Projects and Tasks in Trash** 4a.1 Planner Select Filter Rules 4a.2 System Continues in Step 2
Exception Flow	**2a. No Exists Projects and Tasks in Trash** 2a.1. System displays "Trash is empty" mention in Trash.

11. UC11_Using Real Time Tracker

Use Case Name	View Real Time Tracker
Actor	Planner, Clokify
Description	For Planner to tracker their real time using in the projects or tasks.
Precondition	• Planner must sign in to the application • Planner must create projects or tasks.
Postcondition	• Planner can view the time taken on their projects or tasks
Standard Process	1. Planner opens the projects 2. System displays the project dashboard and task 3. Planner ticks the time tracker's box to open the tracker system 4. System connects and accesses the third-party system as Clockify with using API. 5. System successfully connects and accesses 6. Planner open the hardware, software or system to work 7. Clockify detects the planner start work 8. Clockify starts record the time and take the attendance 9. System recording and copy the Clockify data 10. Planner end the close the hardware, software or system 11. Clockify detects the planner end work 12. Clockify ends to record the time and display the offline time 13. System stores and processes final time logs, making them available for reports, billing, or audit. 14. Planner view the real-time using in the project dashboard
Alternative Path	**10a. Short Time No Motion** 10a.1. Planner no motion on hardware, software or system 10a.2. System stops recording time

	10a.3. System starts the breaktime timer 10a.4. Planner back to use hardware, software, system 10a.5. System stops breaktime timer 10a.6. System continues to step 7. **<u>10b, Long Time No Motion</u>** 10a.1. Planner no motion on hardware, software or system 10a.2. System stops recording time 10a.3. System starts the breaktime timer 10a.4. System timer over break time 10a.5. System continues to step 12.
Exception Flow	**<u>4a. Failed Connect Clockify</u>** 4a.1. System failed to connect Clockify 4a.2. Planner try to restart application or network or check the time tracker has opened or not. **<u>9a. Failed Access Data</u>** 9a.1. System failed to connect Clockify Database or copy data 9a.2. Planner try to restart applications or network.

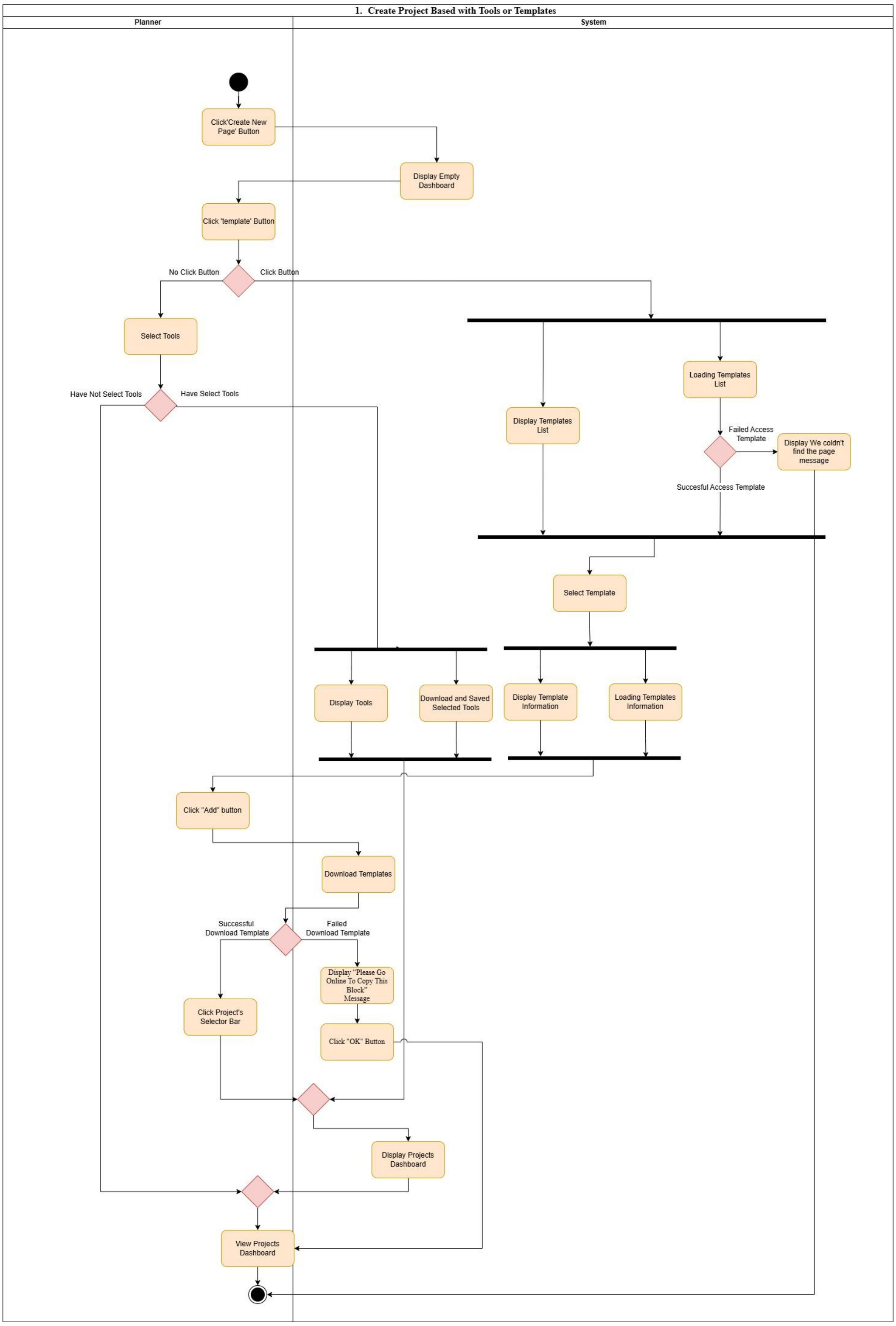
12. UC12_View Visualize Graph

Use Case Name	View Visualize Graph
Actor	Planner
Description	For planner to view the overall status of projects and tasks in visualize graphs
Precondition	• Planner must sign in to the application • Planner must create the projects
Postcondition	• Planner can view the visualize graph
Main Flow	1. Planner opens the project dashboard 2. System displays the project dashboard. 3. System frontend request data by using API 4. System loading and analysis overall of the project and tasks 5. System generates visualize graph 6. System embeds visualize graph on project dashboard 7. Planner can view the visualize graph on project dashboard
Alternative path	-
Exception Flow	**<u>3a. Failed to Connect API</u>** 3a.1 System failed to connect API 3a.2. System records the error details

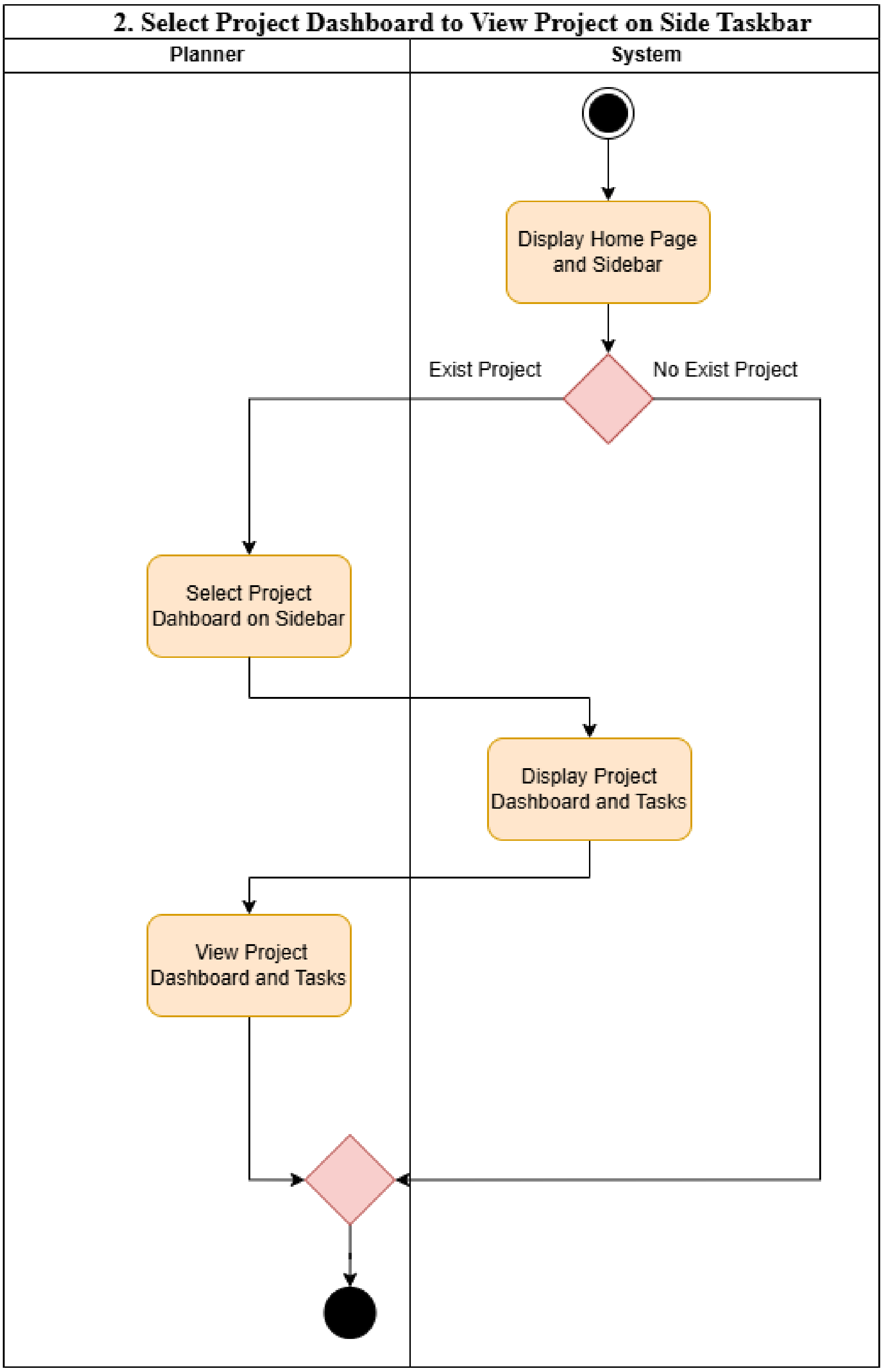
	3a.3. System displays "Failed to load project data" message. 3a.4. Planner close the application or check network to try again **<u>4a. Failed to Access Data</u>** 4a.1. System failed to access the data 4a.2. System record data error quantity and reasons 4a.3. System creates a list or error data 4a.4. System display the error of list data

5.0 Activity Diagram

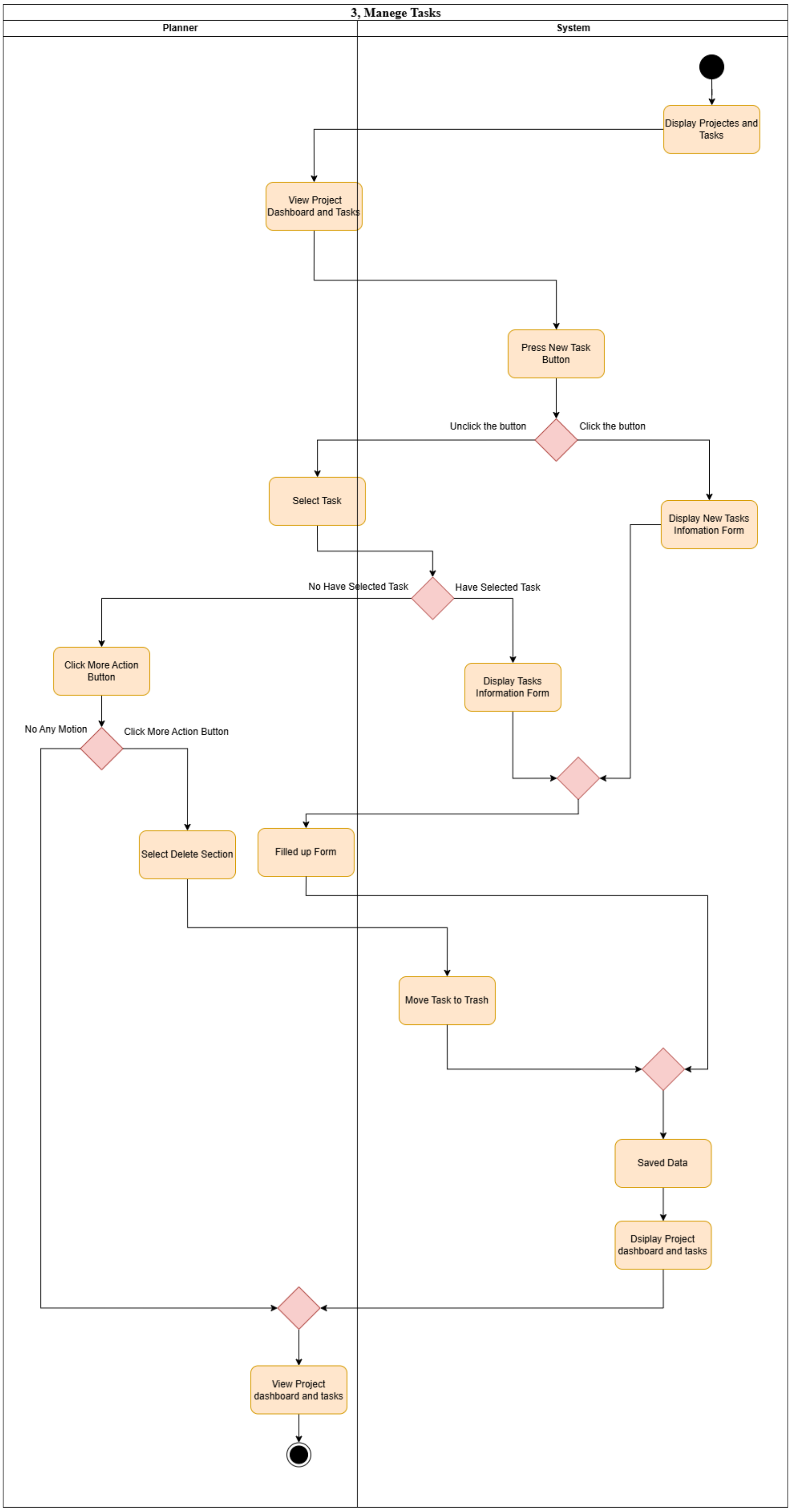
1. Create Project based with Tools or Templates



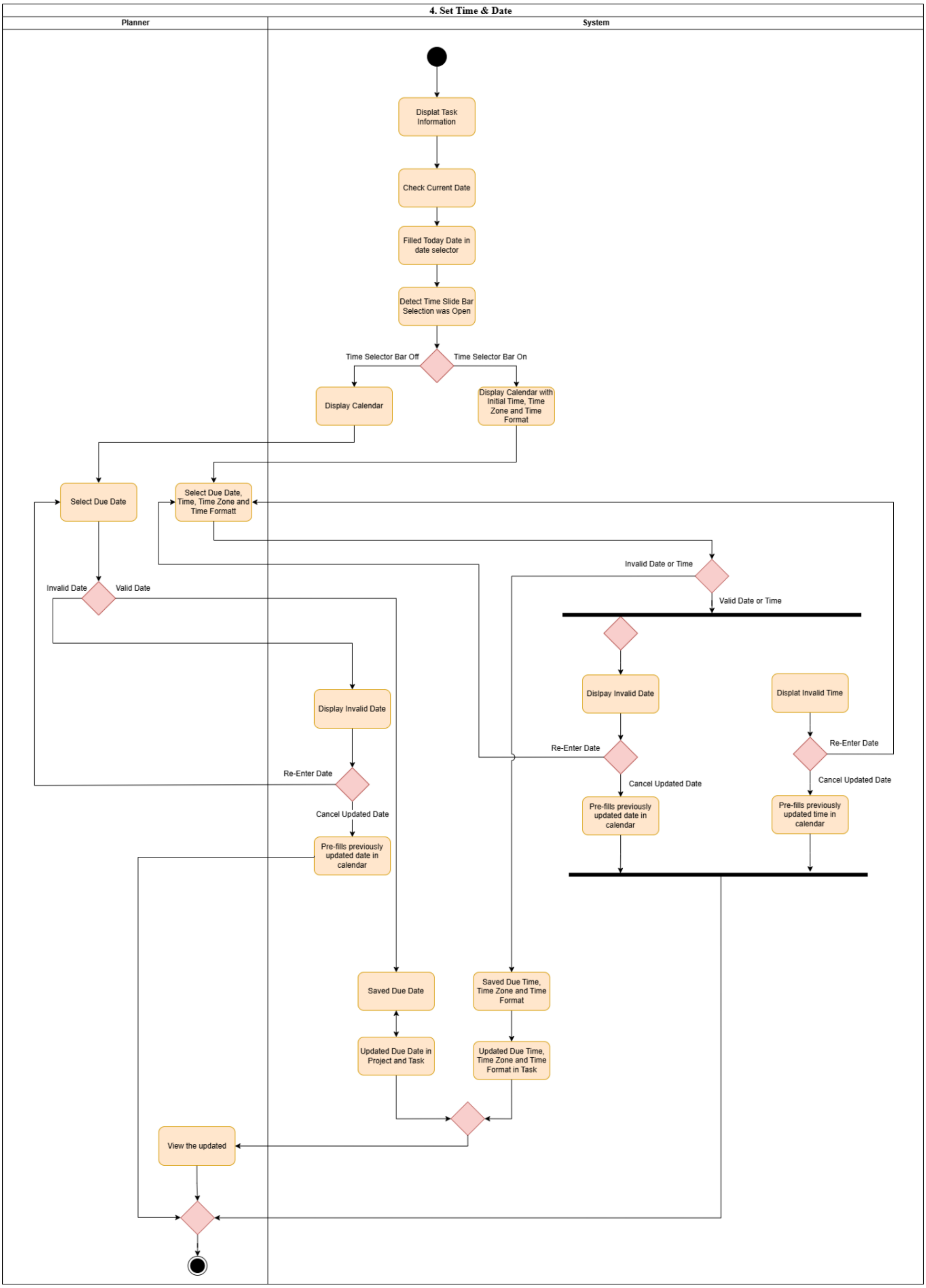
2. Select Workspace to View Project on Side Taskbar



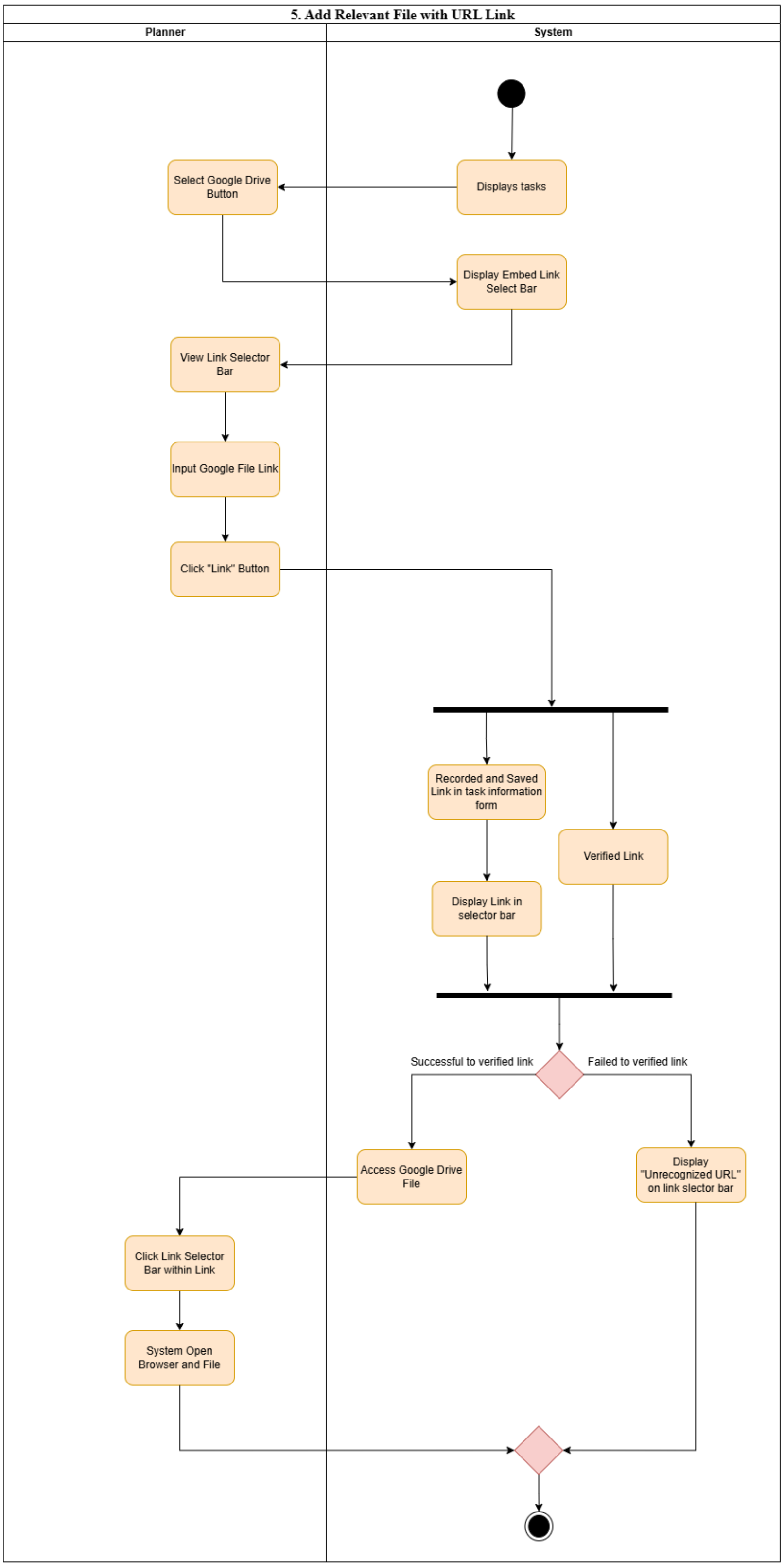
3. Manage Task



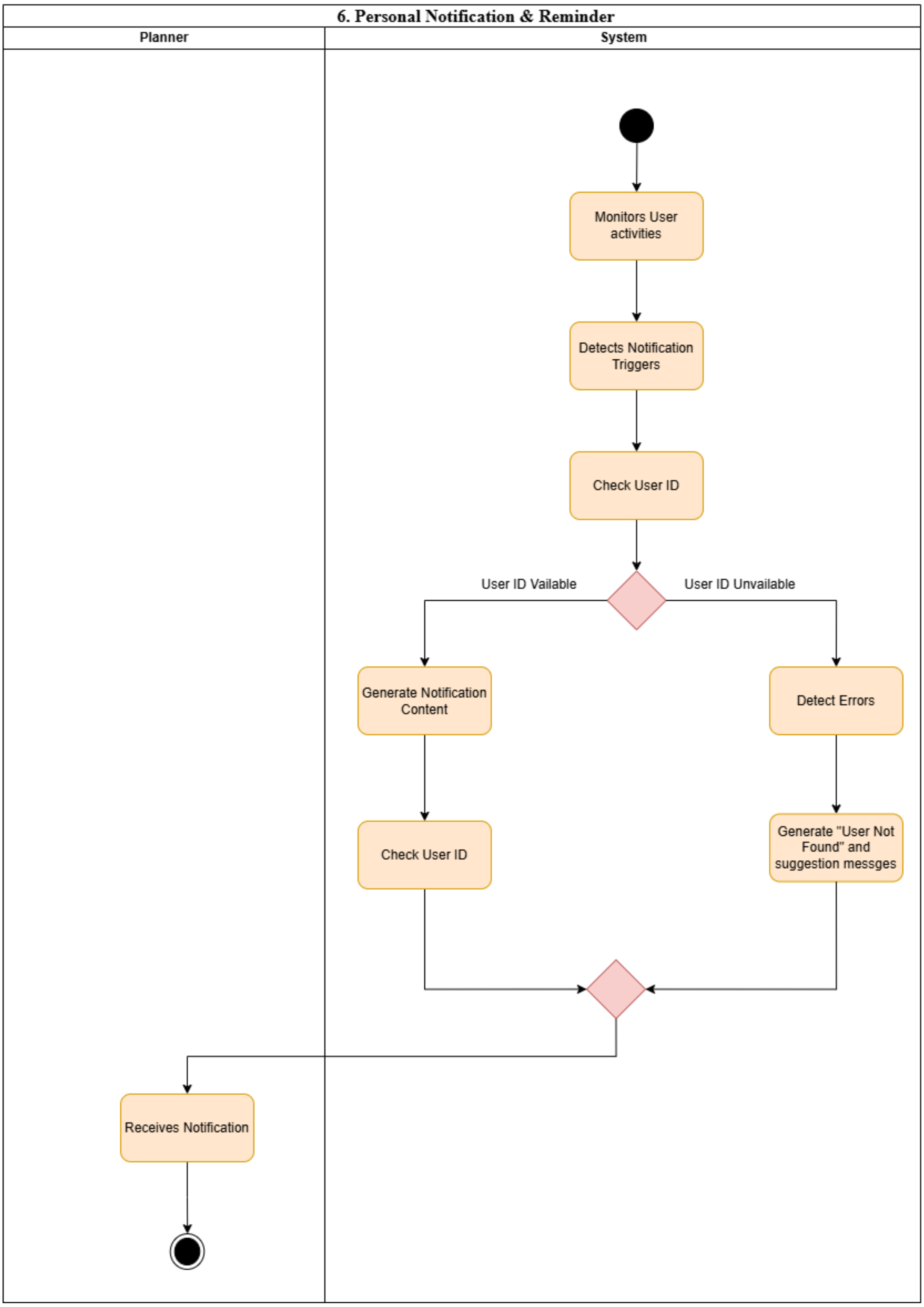
4. Set Time & Date



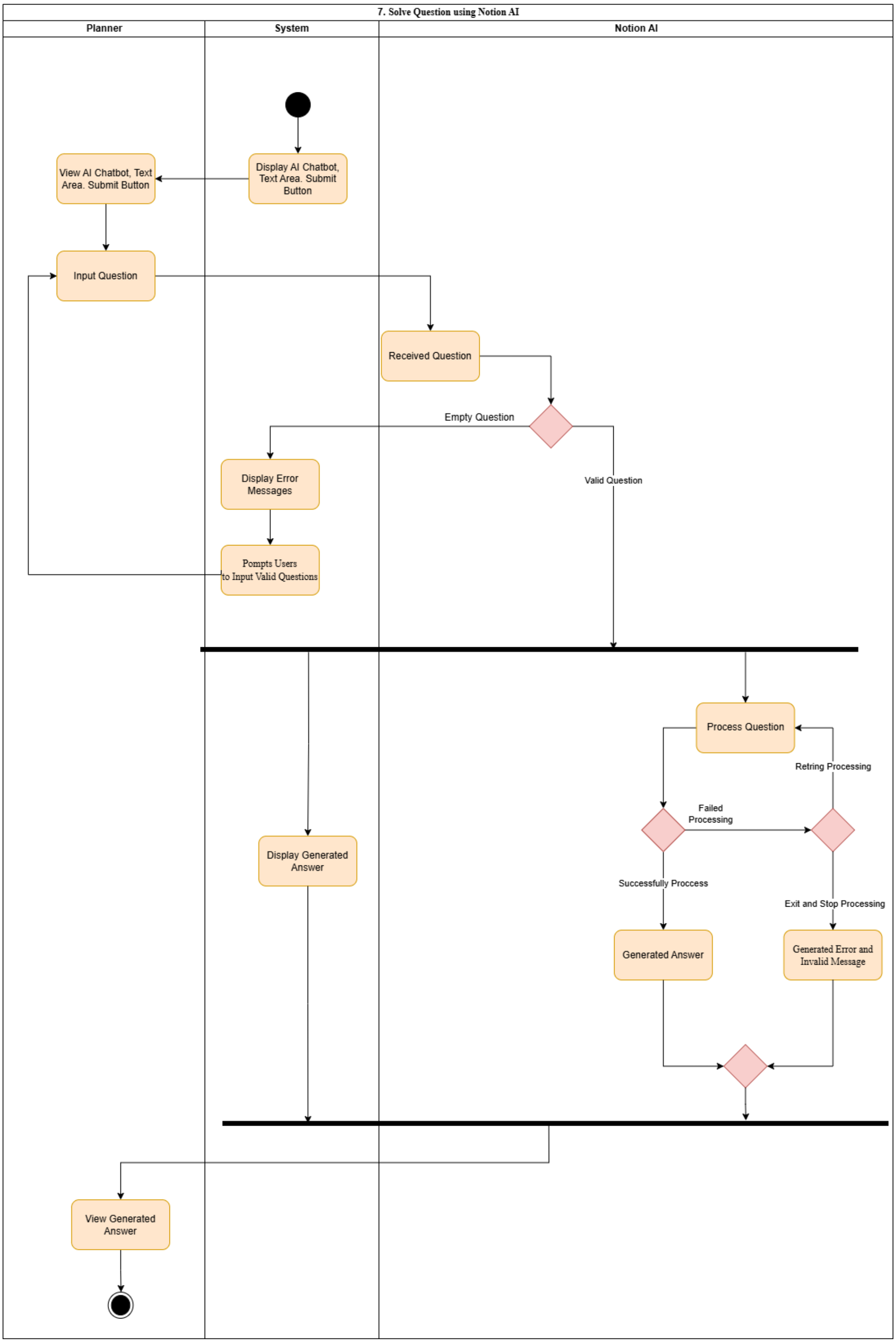
5. Add Relevant File with URL Link



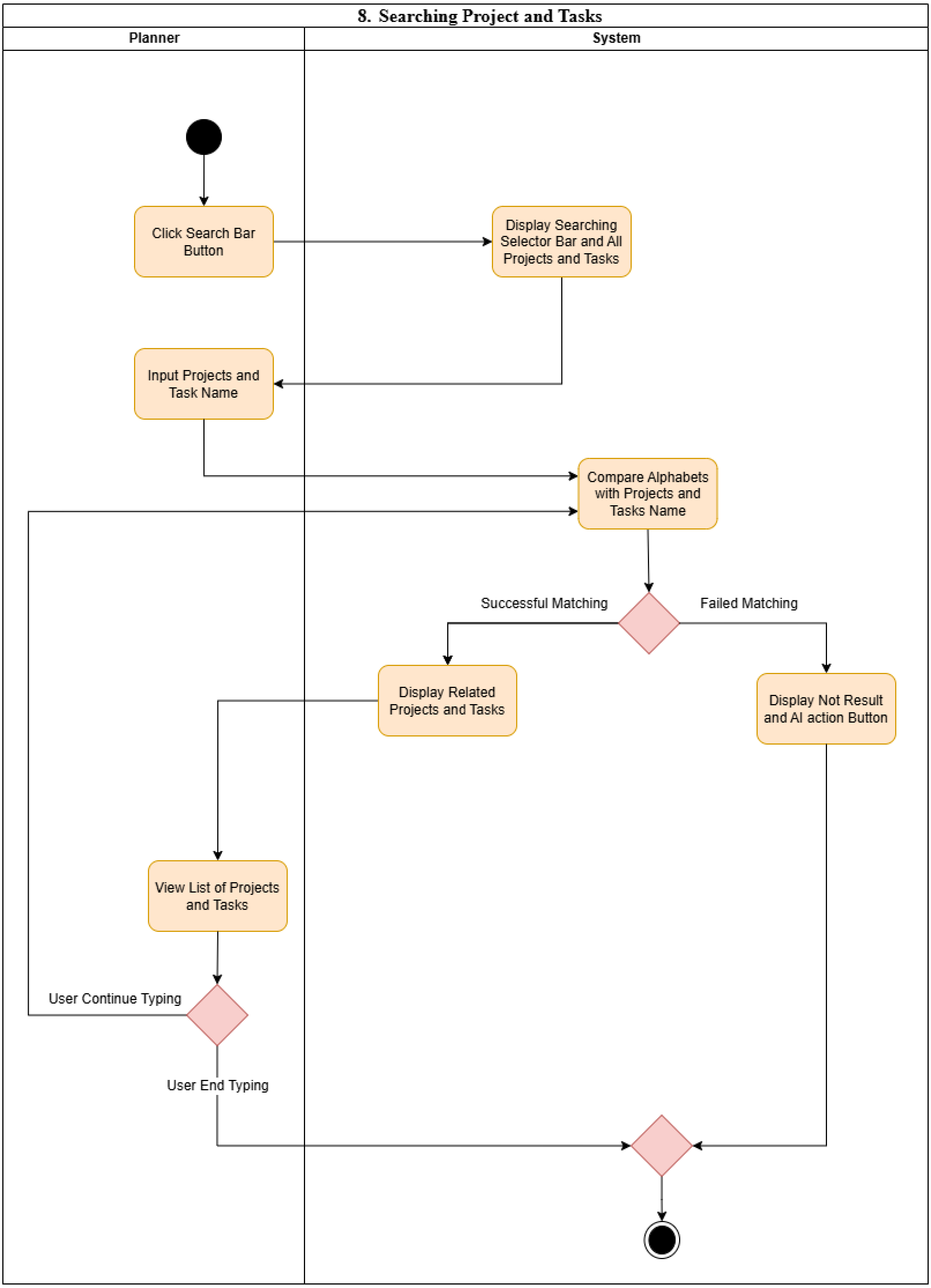
6. Personal Notification and Reminder



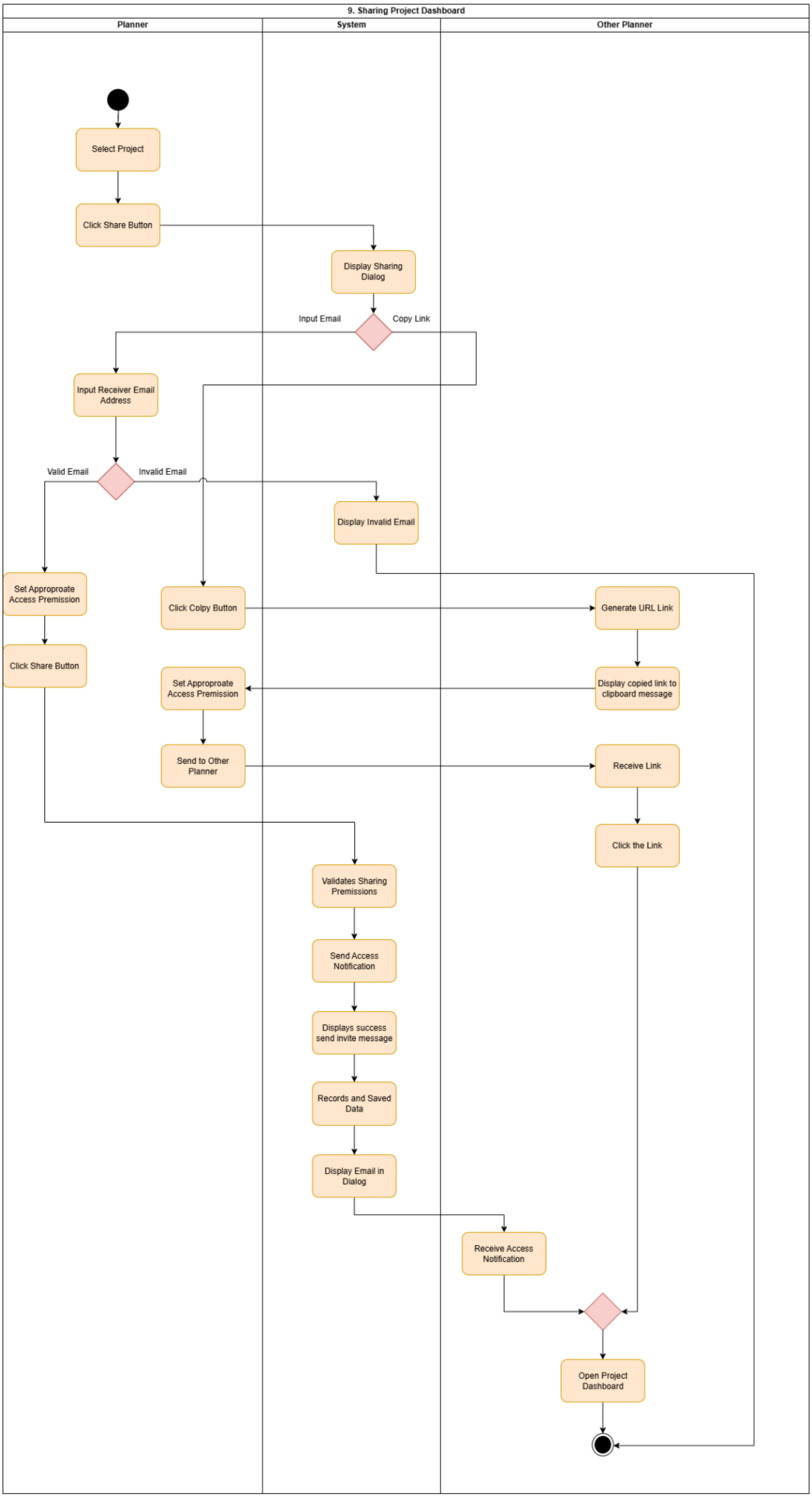
7. Solve Questions Using Notion AI



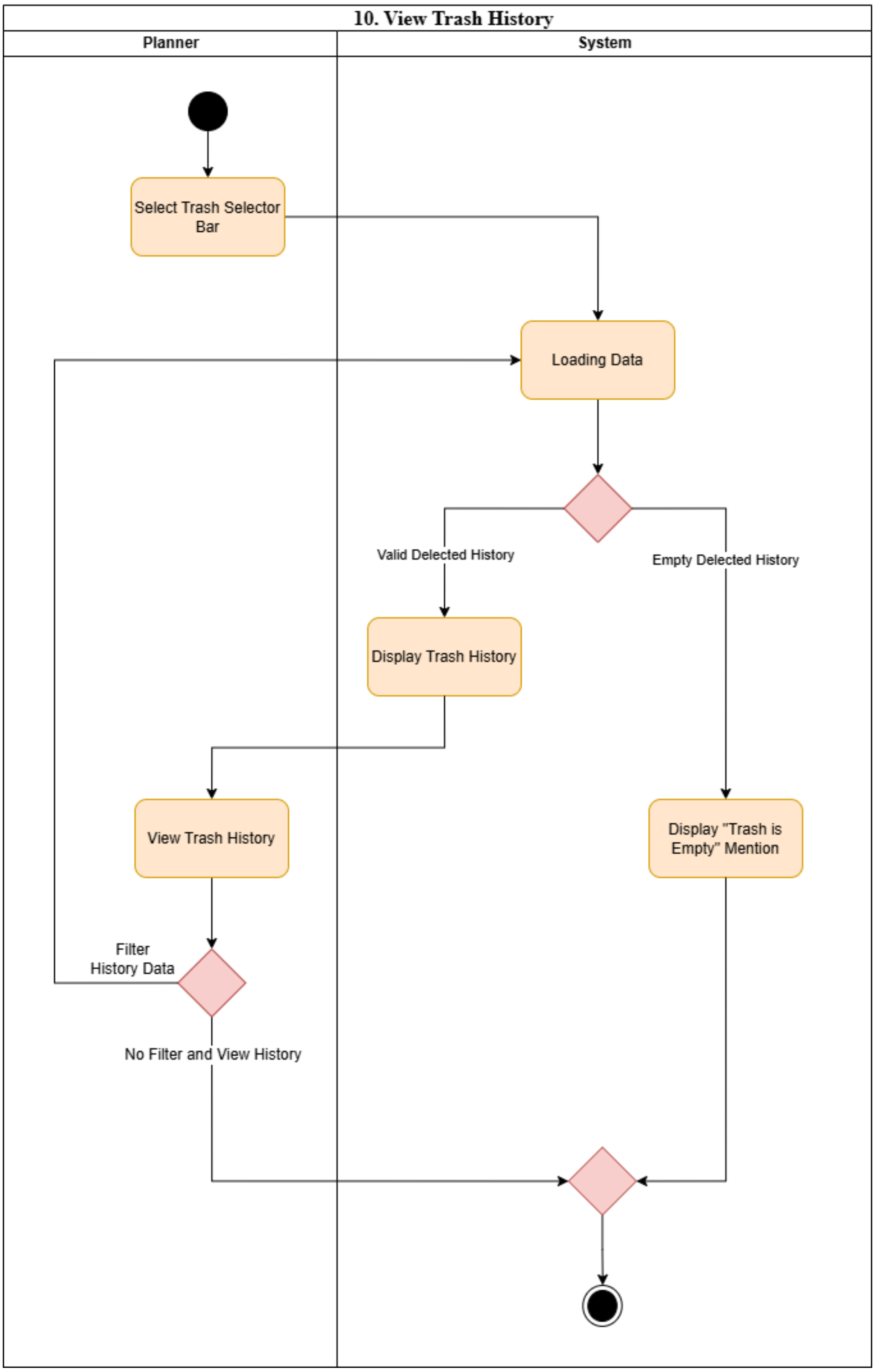
8. Searching Projects and Tasks



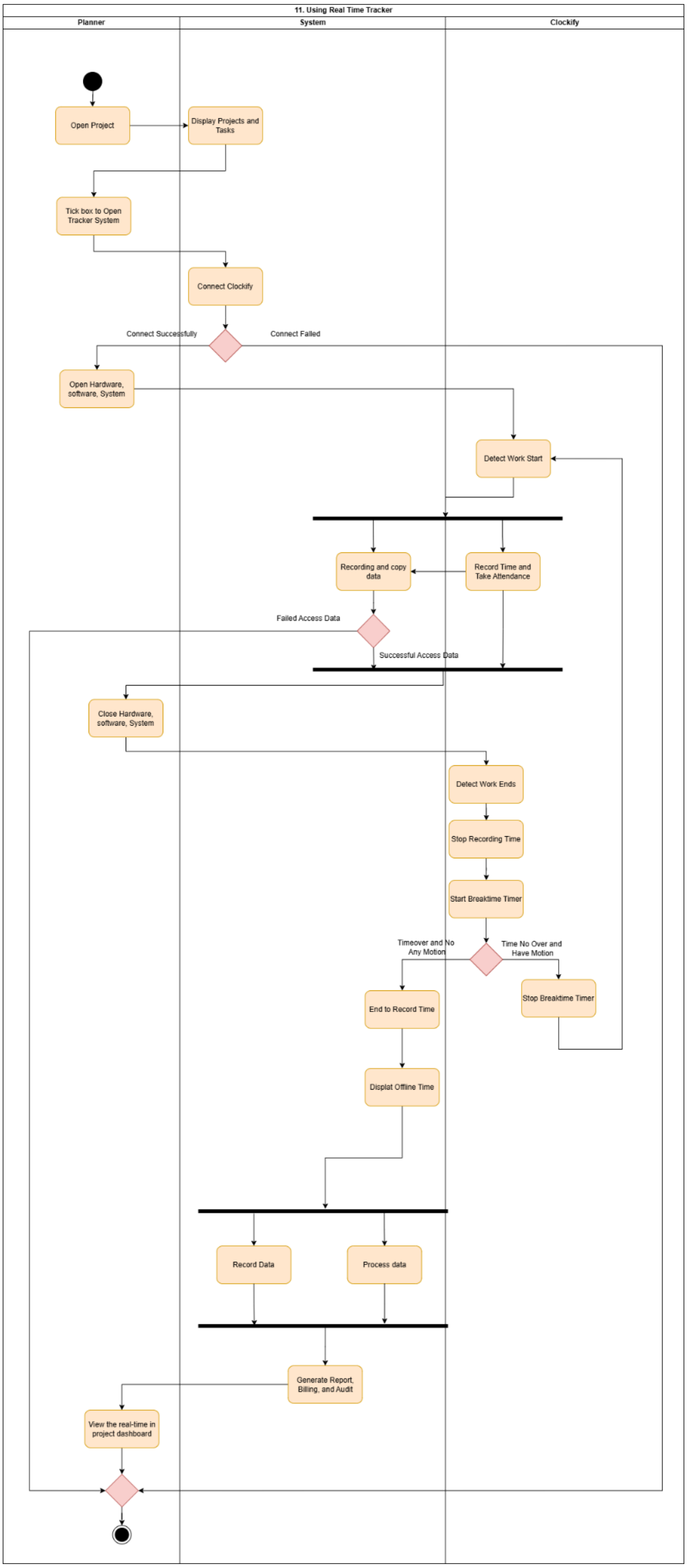
9. Sharing Project Dashboard



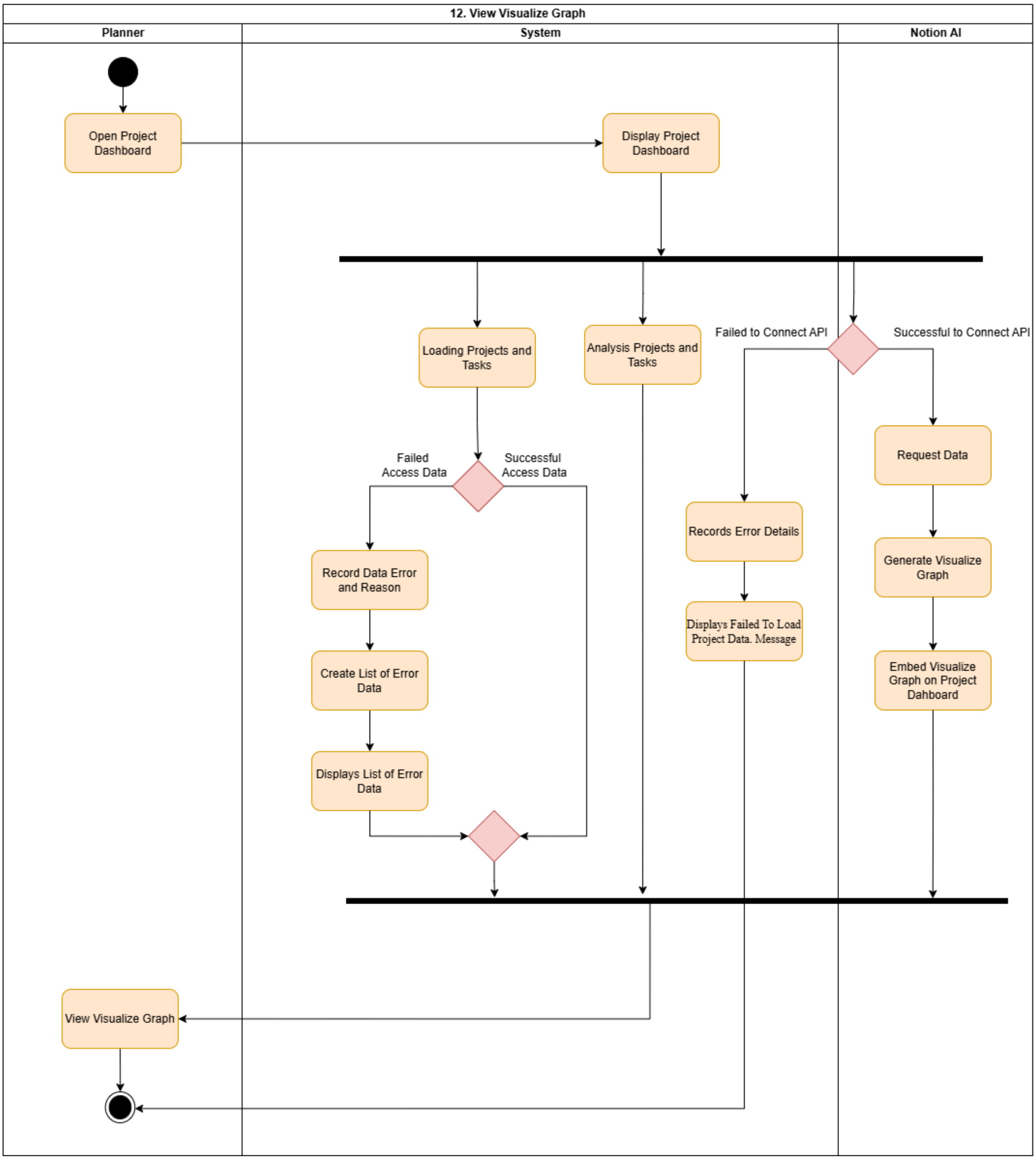
10. View Trash History



11. View Real Time Tracker



12. View Visualize Graph



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CT015-3-2

DESIGN METHODS

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Table of Contents

Table Content of Figure

1.0 Design Model

1.1 Class Diagram

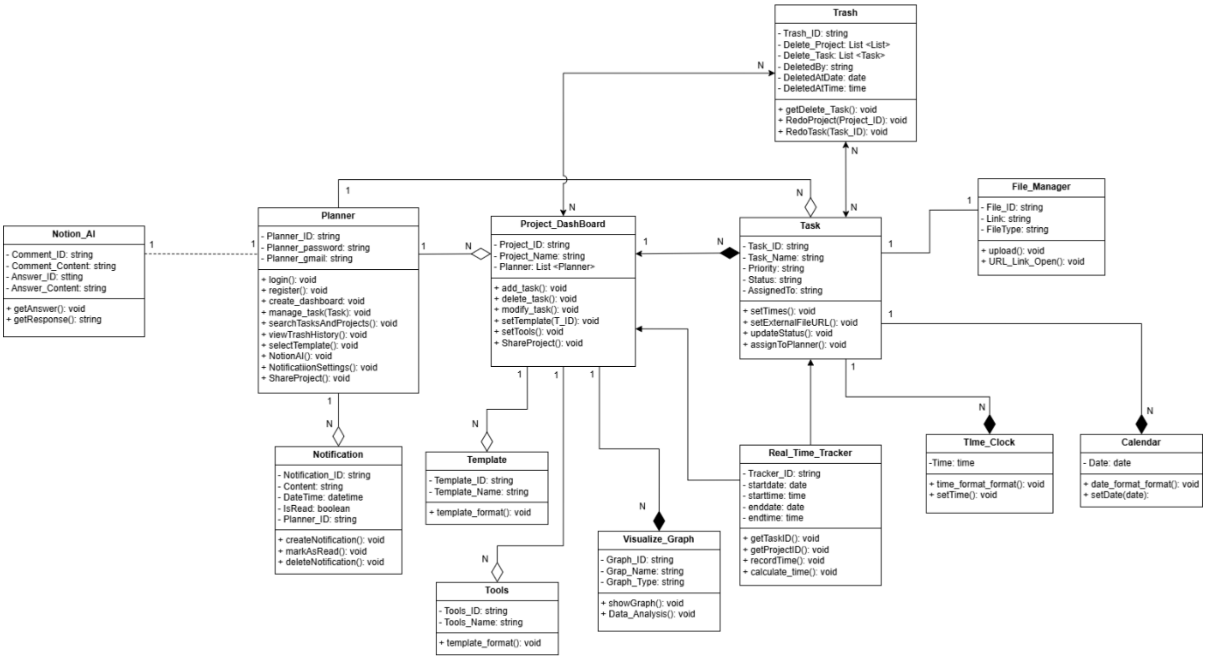

Figure 1 Class Diagram

1.2 Sequence Diagram

1. UC1_Create Project based with Templates or Tools

UC1_Create Project based with Templates or Tools

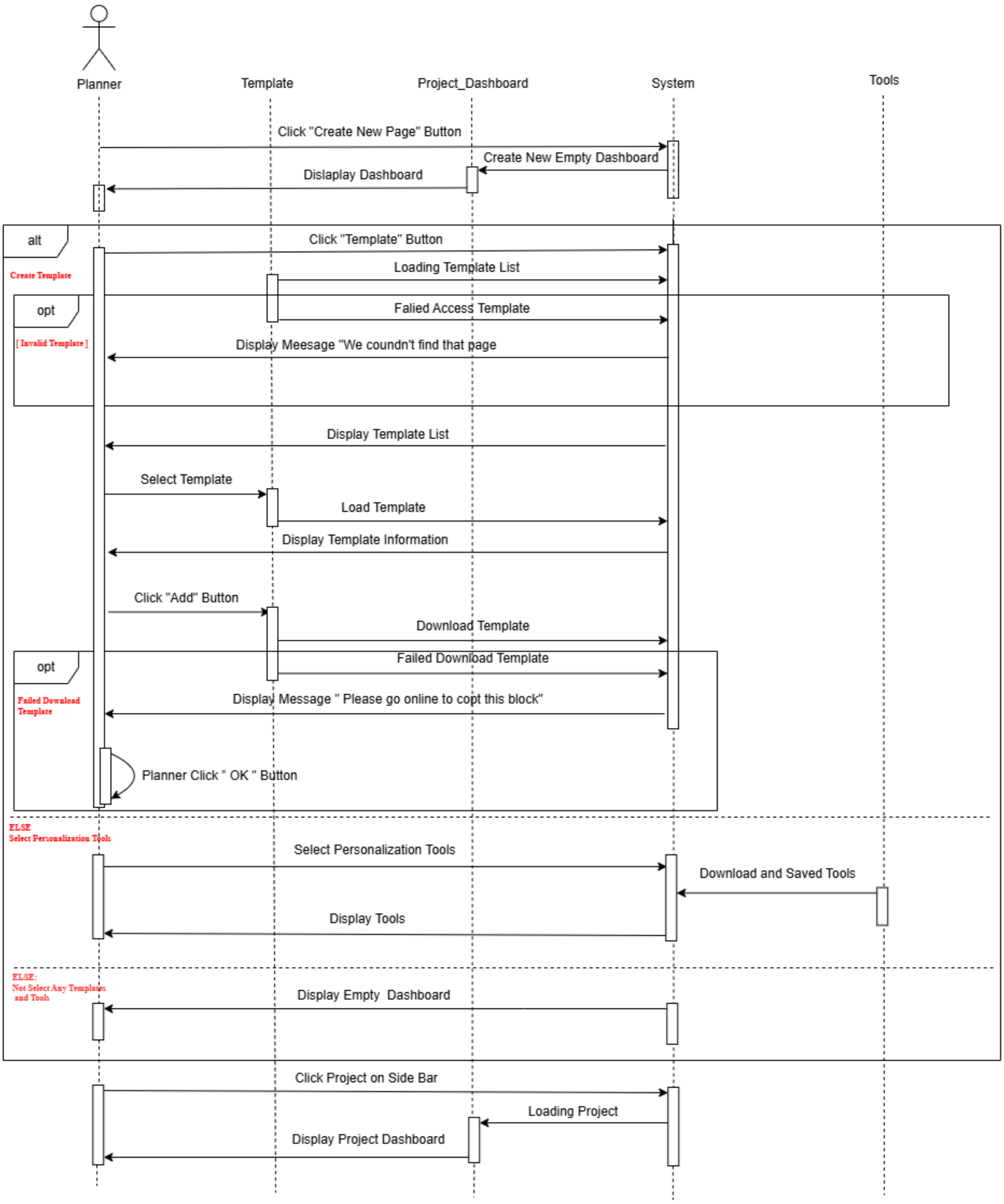

Figure 2 .UC1_Create Project based with Templates or Tools (Sequence Diagram)

2. UC3_Manage Task

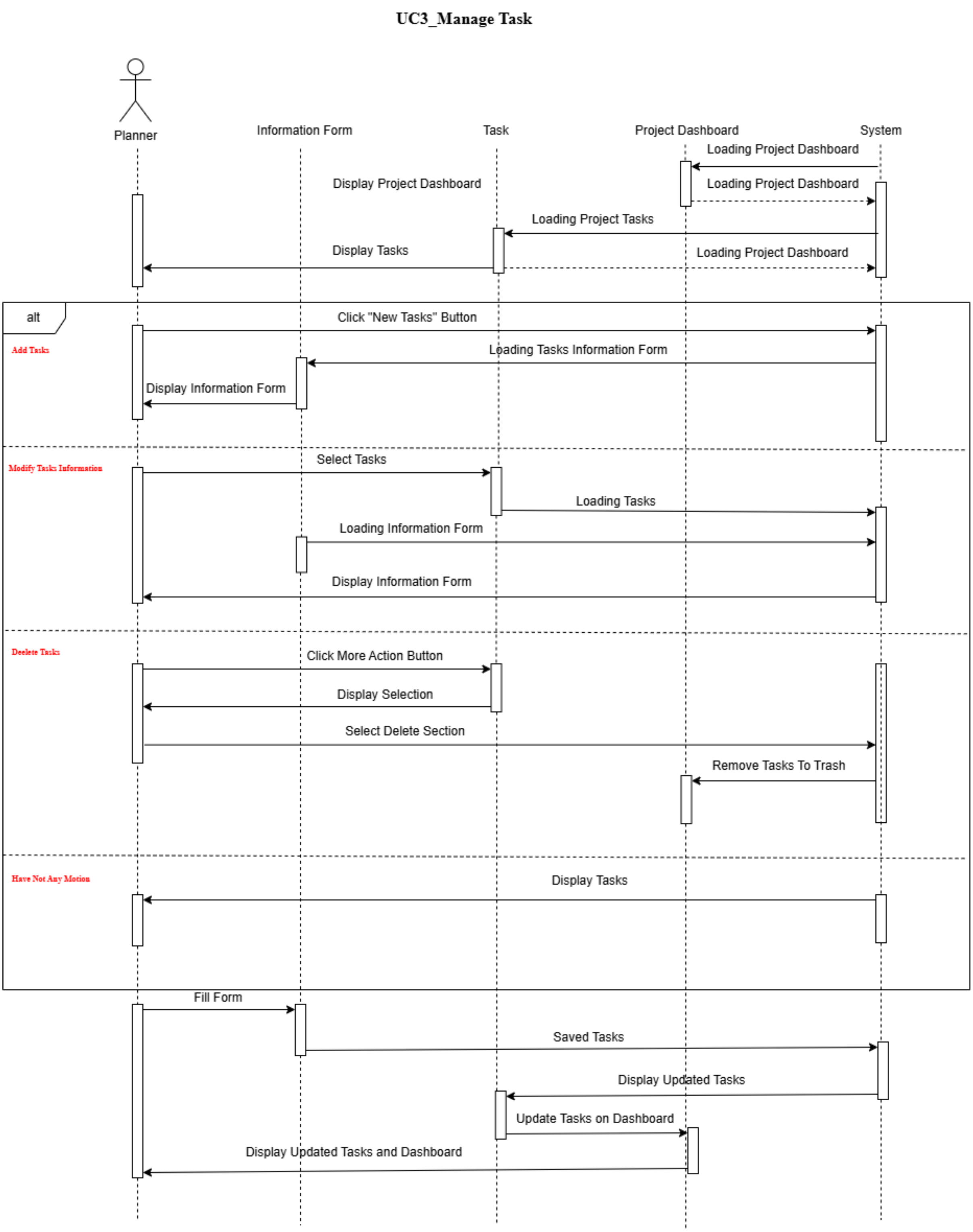

Figure 3 UC3_Manage Task (Sequence Diagram)

3. UC5_Open Relevant File with URL Link

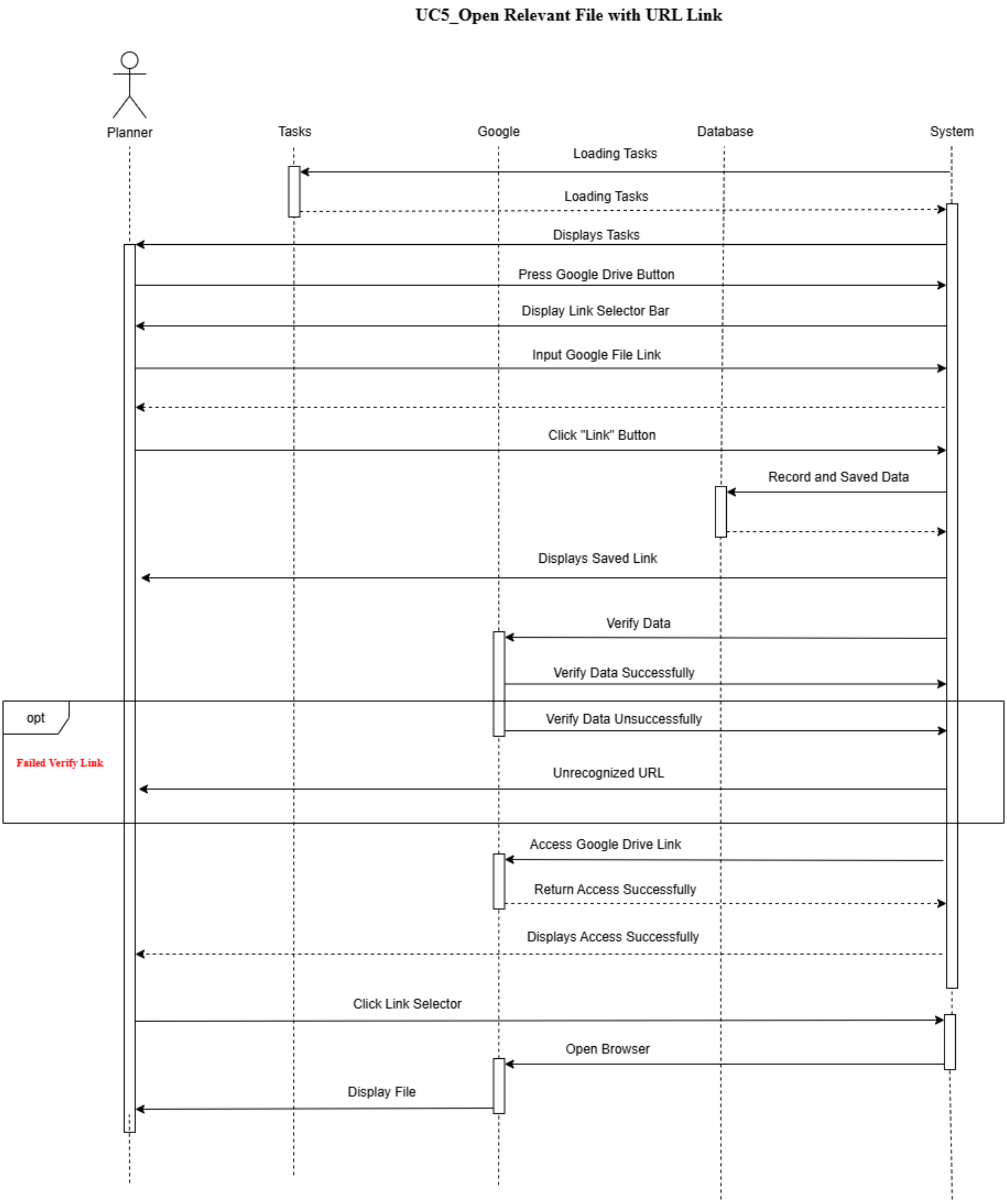

Figure 4 UC5_Open Relevant File with URL Link (Sequence Diagram)

4. UC7_Solve Questions Using Notion AI

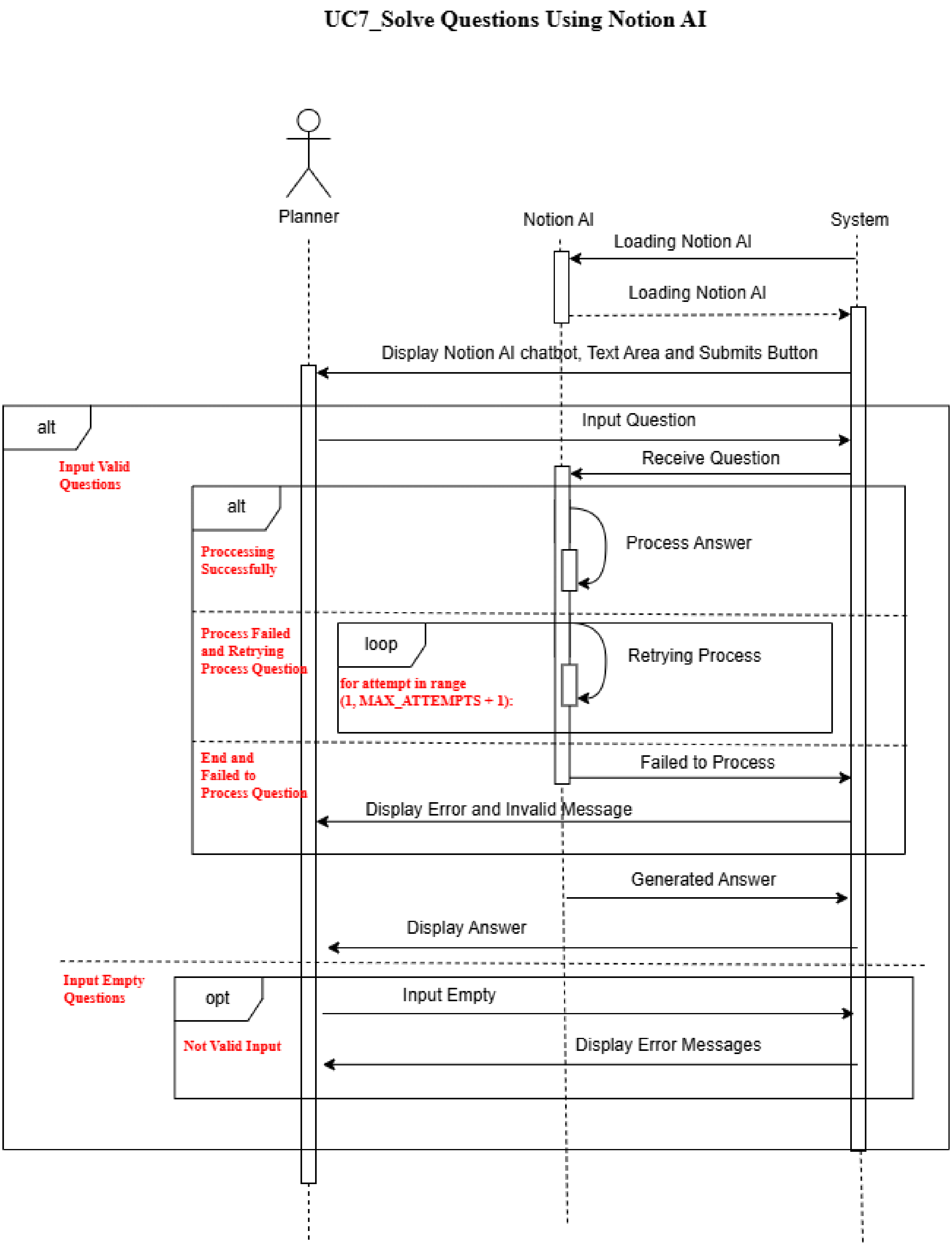

Figure 5 UC7_Solve Questions Using Notion AI (Sequence Diagram)

5. UC8_Searching Project and Task

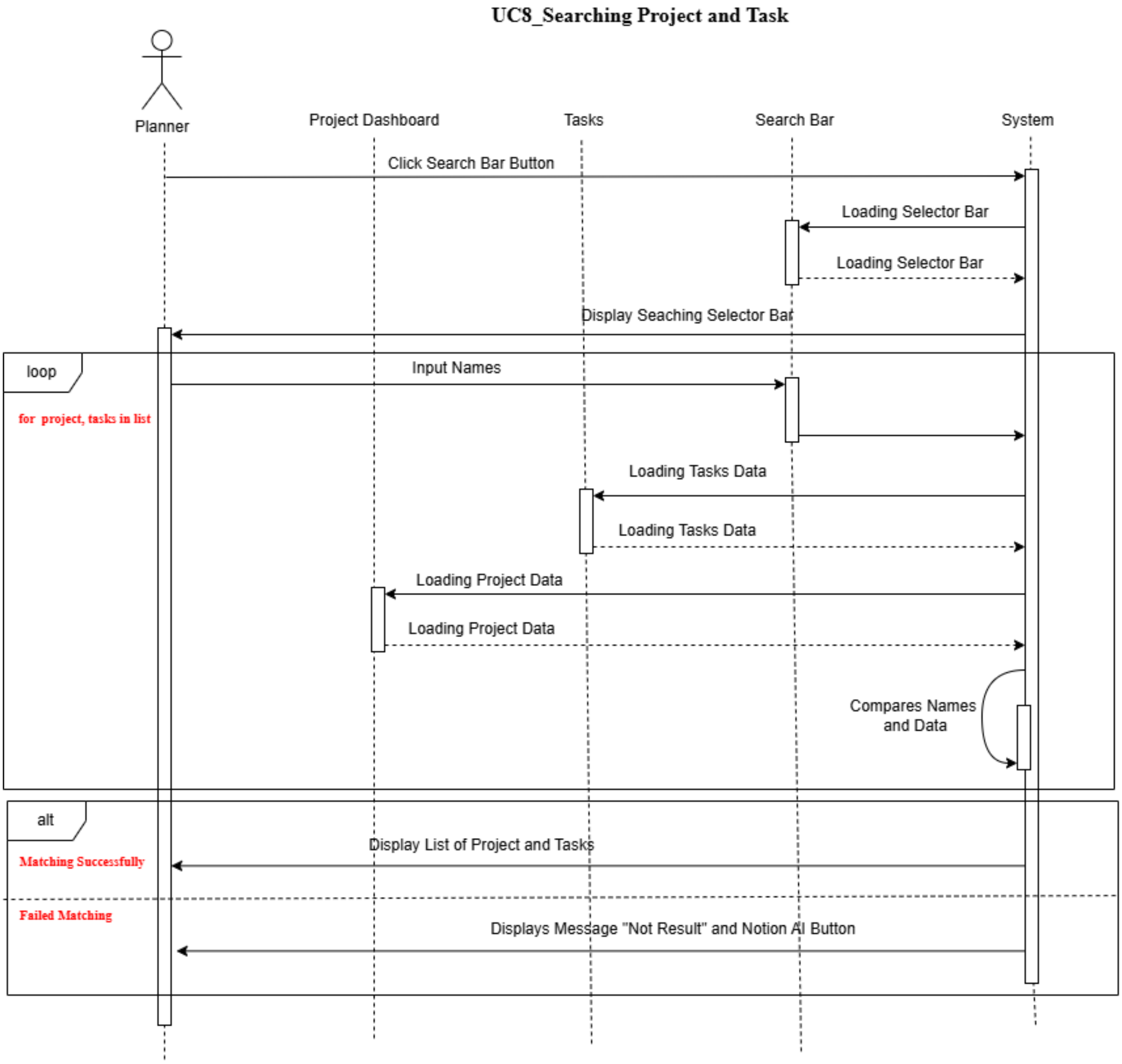

Figure 6 UC8_Searching Project and Tasks (Sequence Diagram)

6. UC10_View Trash History

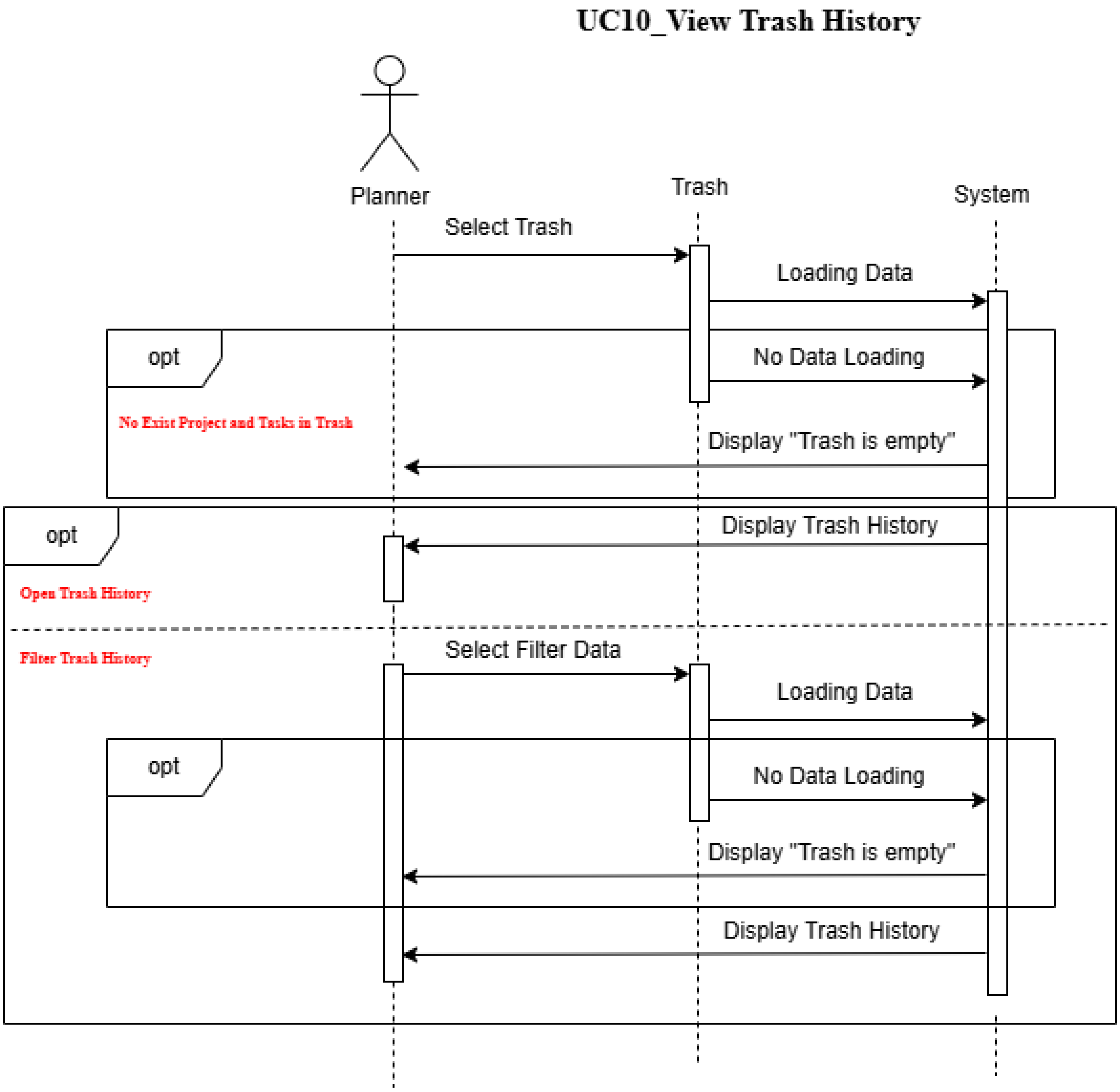

Figure 7 UC10_View Trash History (Sequence Diagram)

7. UC11_Using Real Time Tracker

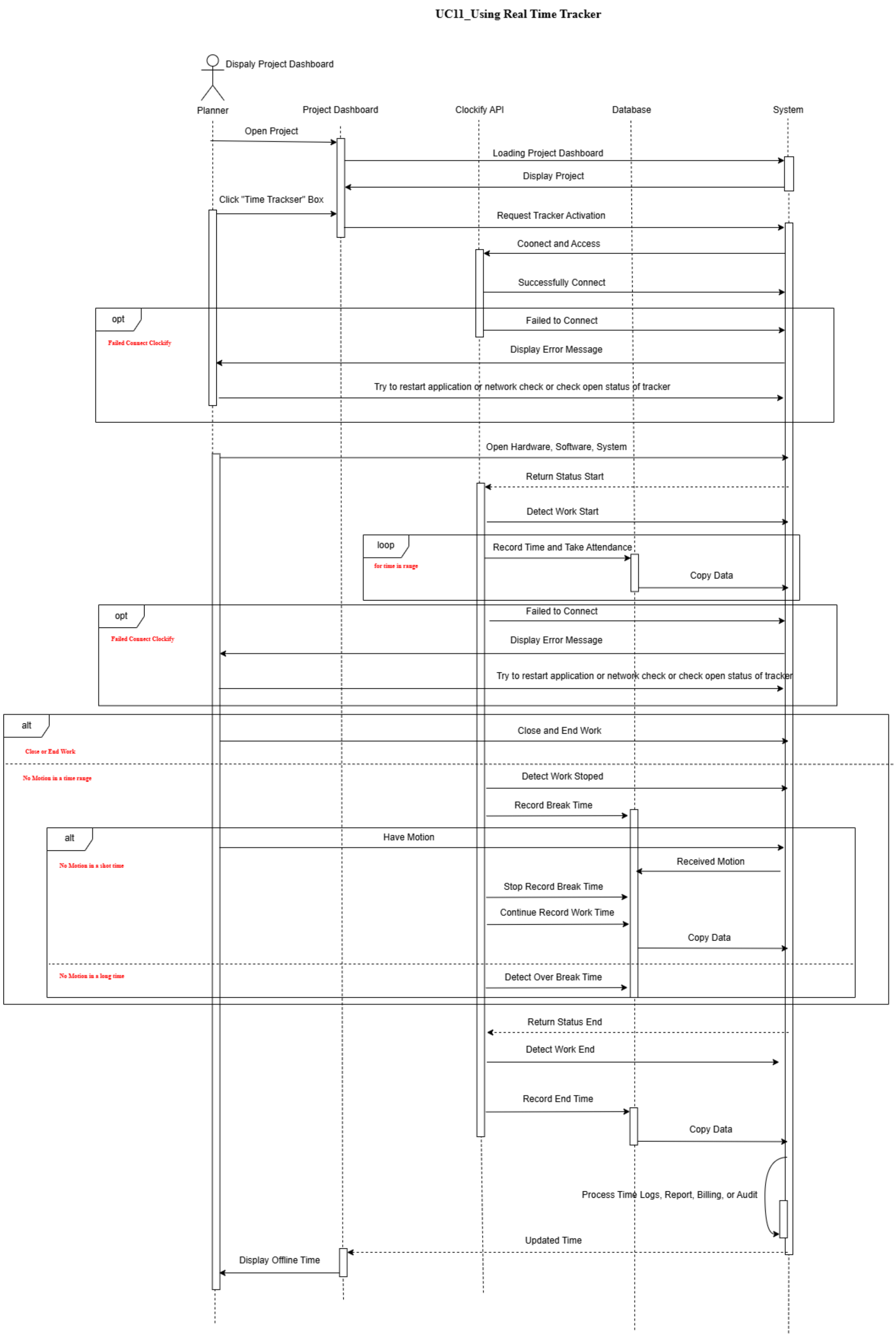

Figure 8 UC11_Using Real Time Tracker (Sequence Diagram)

8. UC12_View Visualize Graph

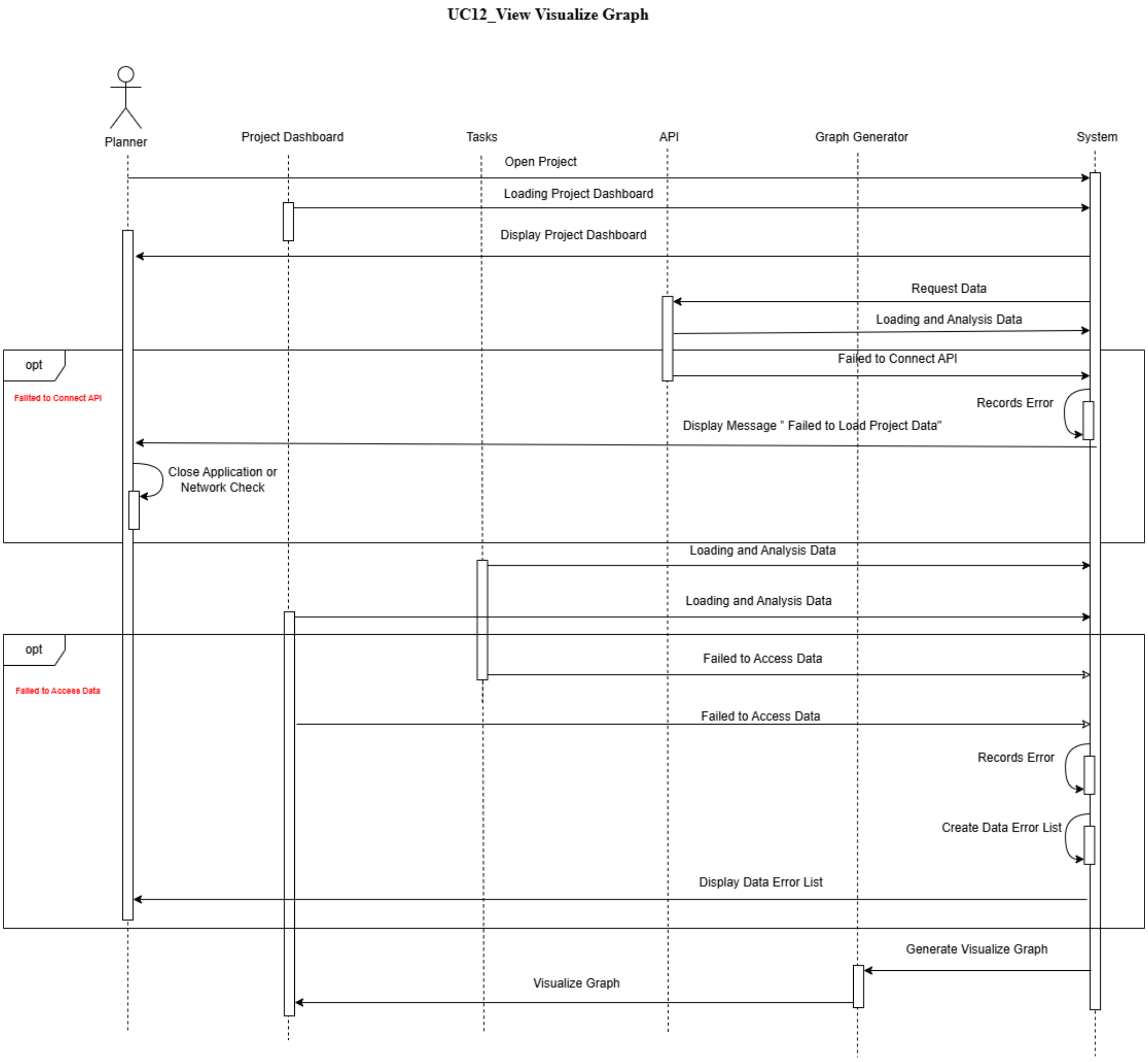

Figure 9 UC12_View Visualize Graph (Sequence Diagram)

1.3 State Machine Diagram

1. UC1_Create Project based with Templates or Tools

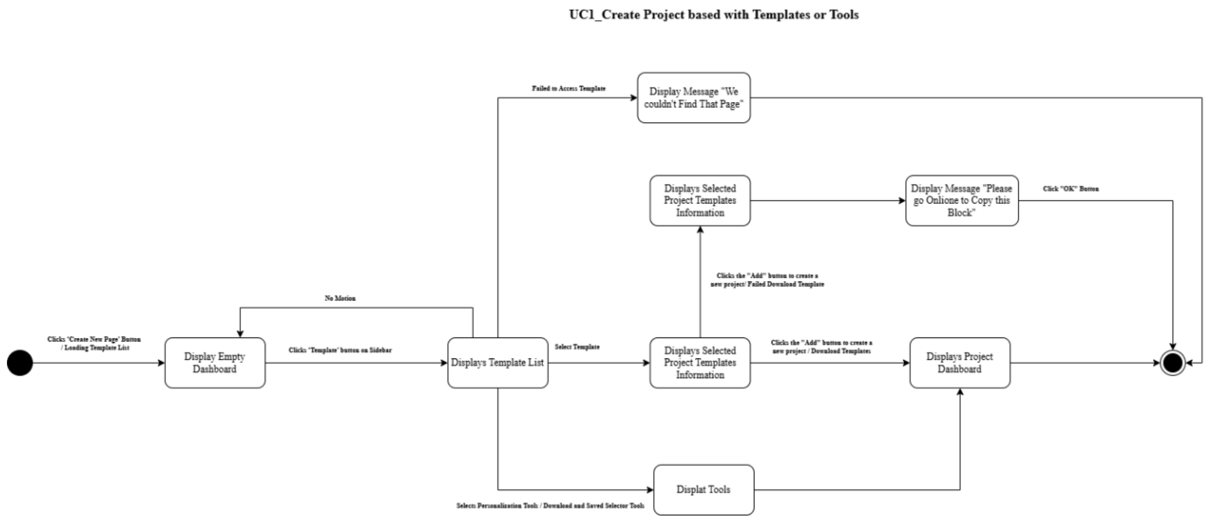

Figure 10 UC1_Create Project based with Templates or Tools (State Machine Diagram)

2. UC3_Manage Task

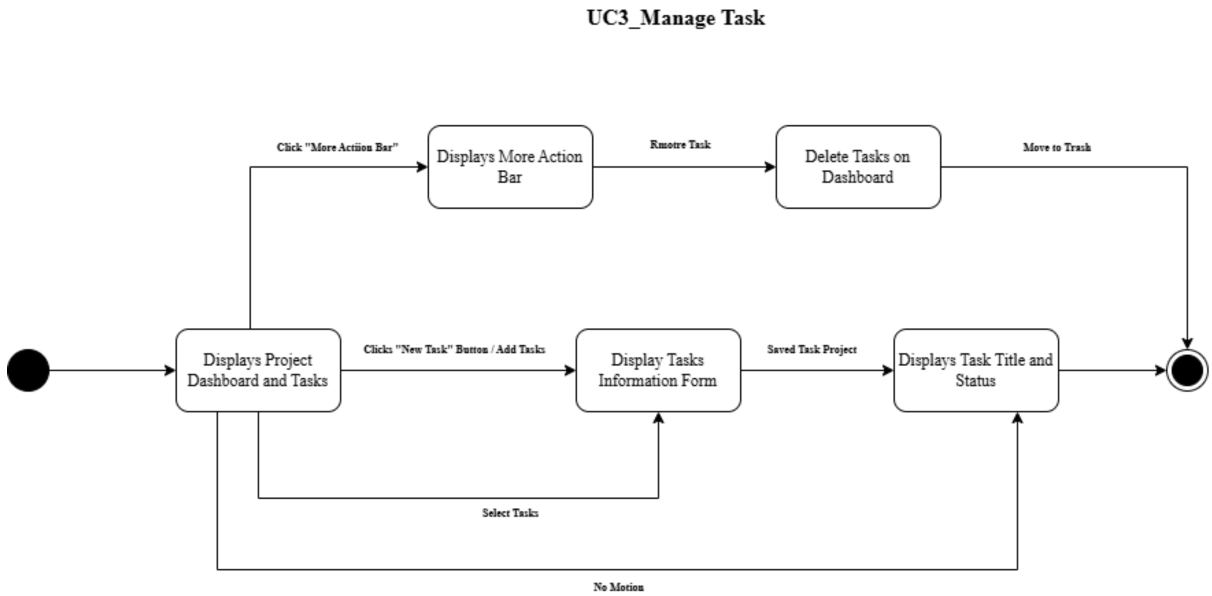

Figure 11 UC3_Manage Task (State Machine Diagram)

3. UC5_Open Relevant File with URL Link

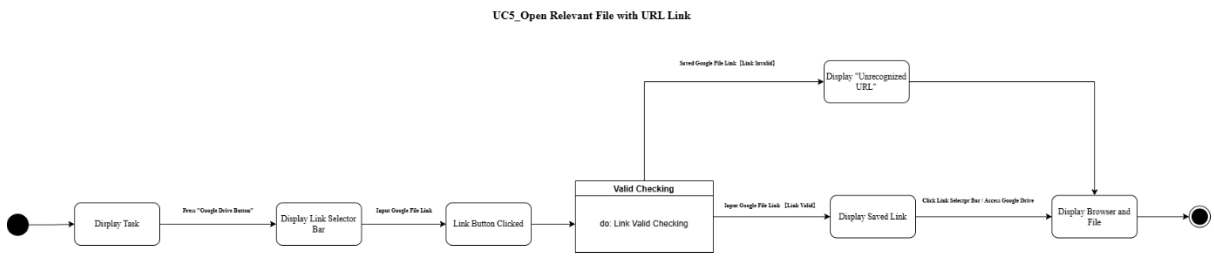

Figure 12 UC5_Open Relevant File with URL Link (State Machine Diagram)

4. UC7_Solve Questions Using Notion AI

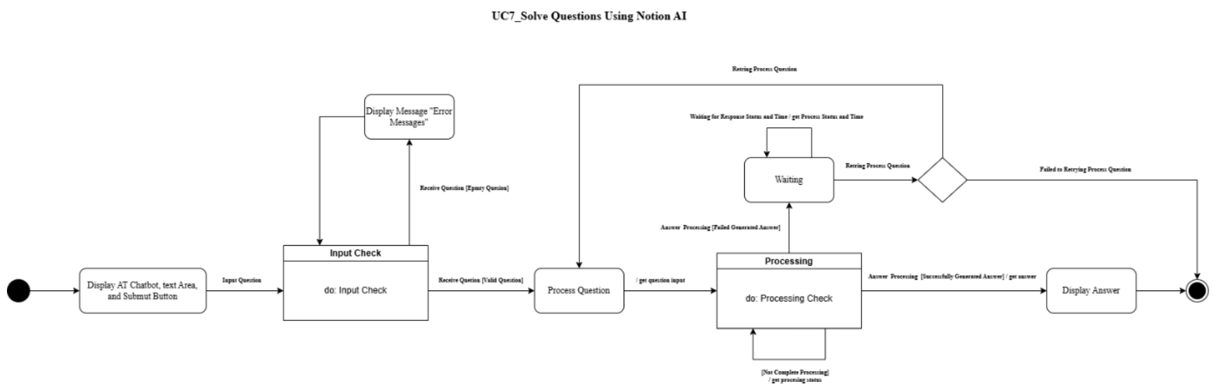

Figure 13 UC7_Solve Questions Using Notion AI (State Machine Diagram)

5. UC8_Searching Project and Task

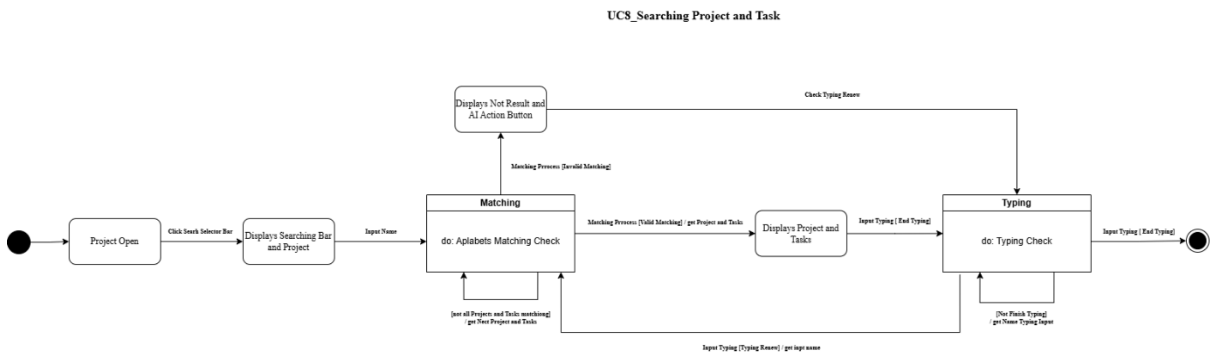

Figure 14 UC8_Searching Project and Task (State Machine Diagram)

6. UC10_View Trash History

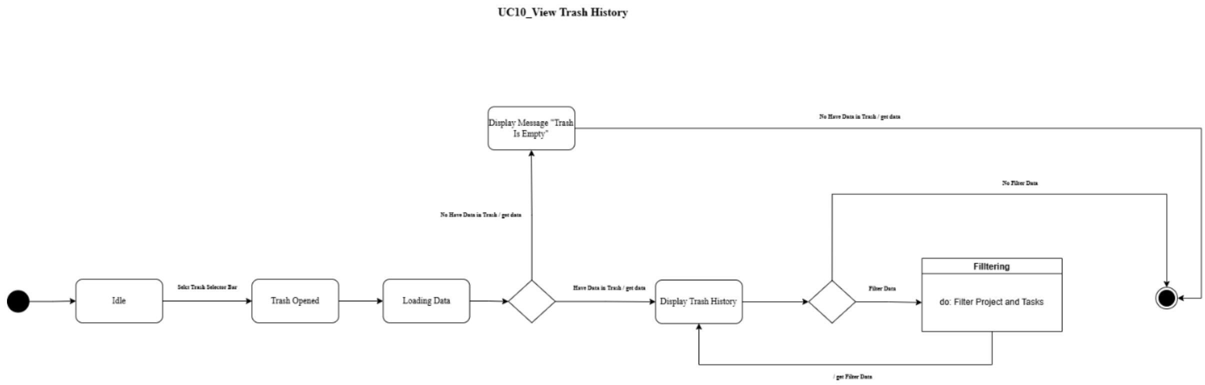

Figure 15 UC10_View Trash History (State Machine Diagram)

7. UC11_Using Real Time Tracker

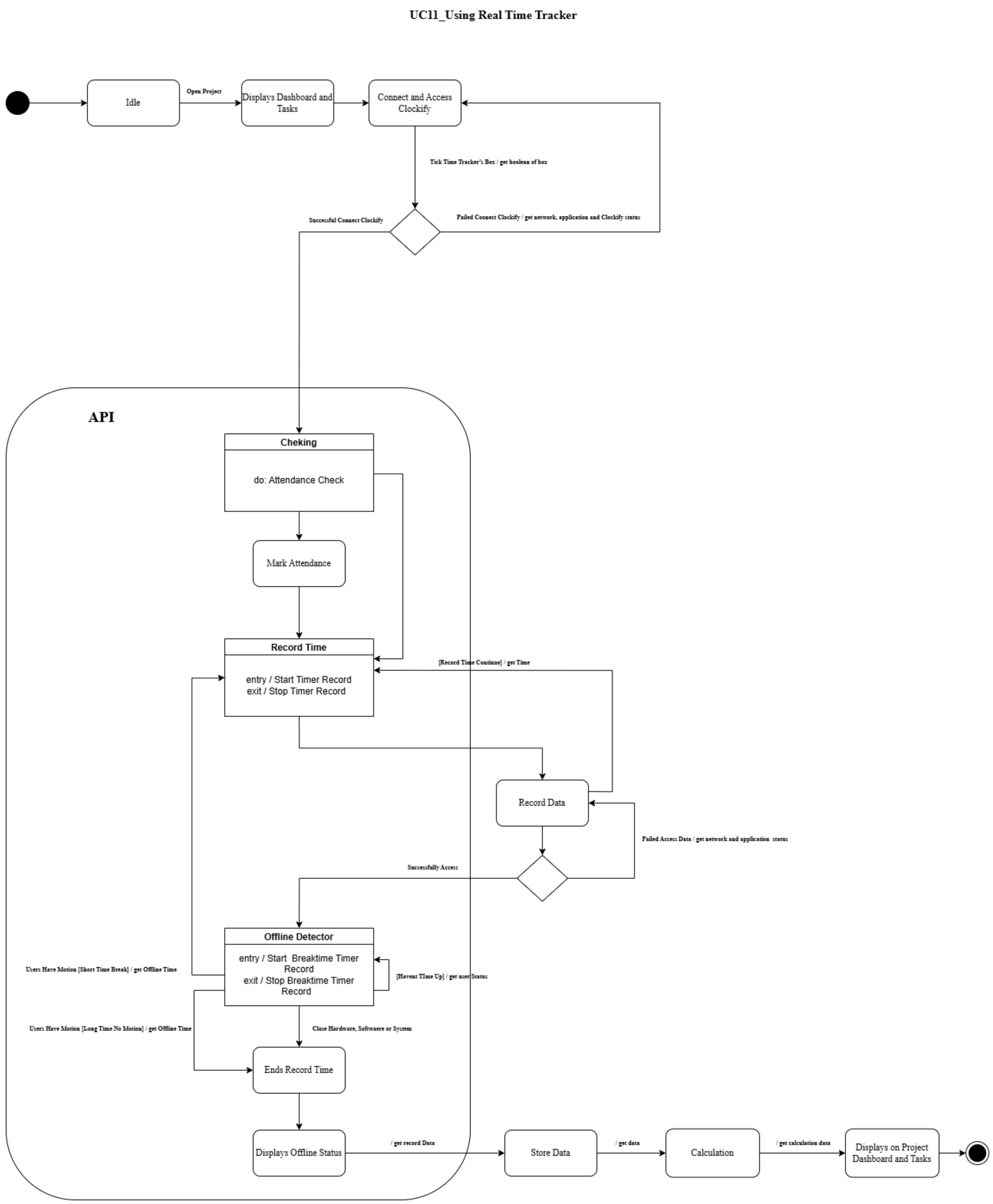

Figure 16 UC11_Using Real Time Tracker (State Machine Diagram)

8. UC12_View Visualize Graph

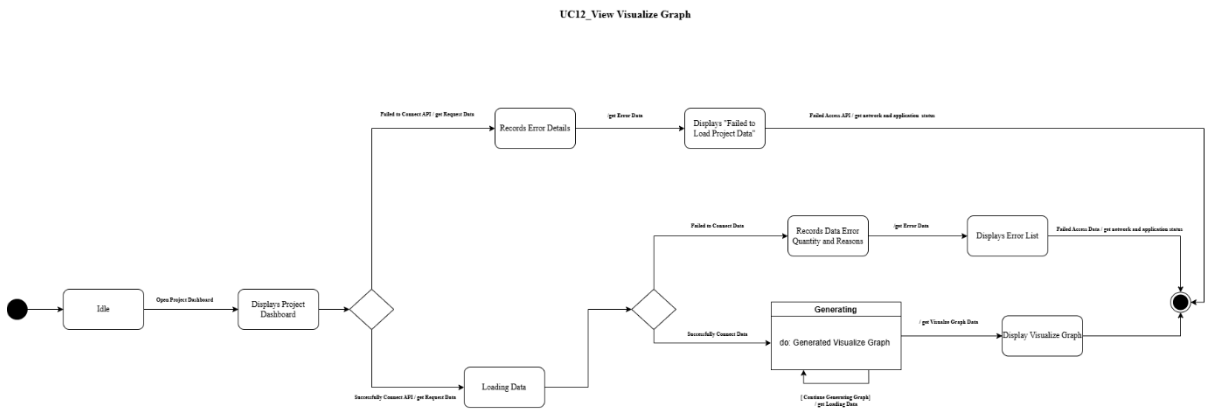

Figure 17 UC12_View Visualize Graph (State Machine Diagram)

2.0 Additional Diagram

2.1 Object Diagram

1. UC1_Create Project based with Templates or Tools

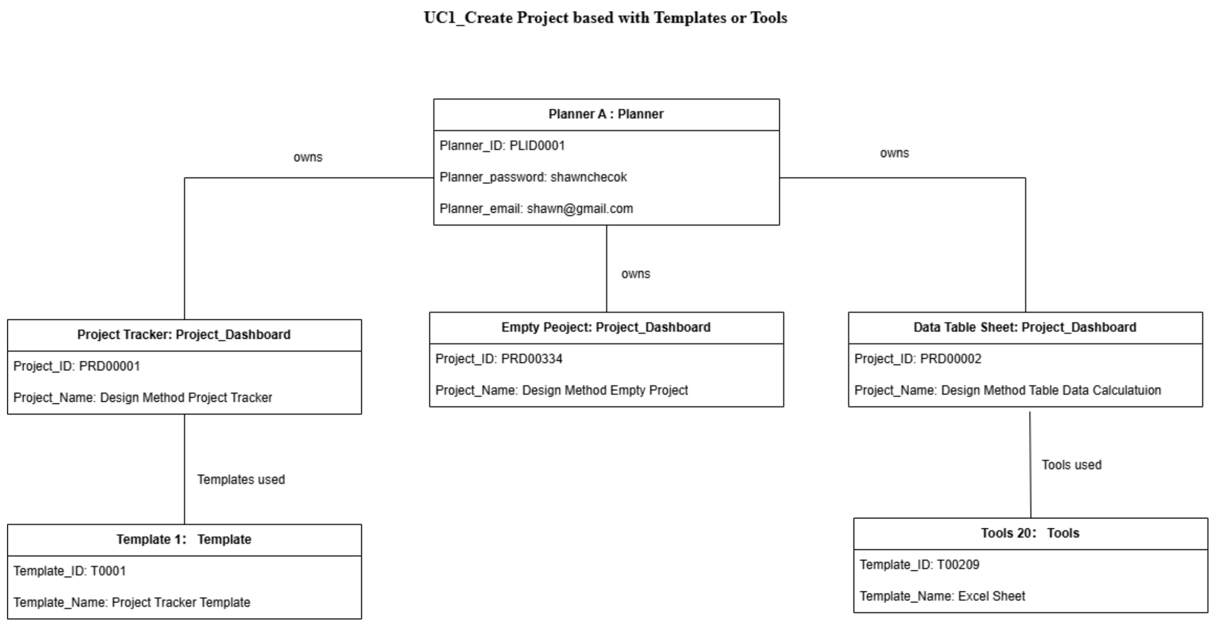

Figure 18 UC1_Create Project based with Templates or Tools (Objective Diagram)

2. UC12_View Visualize Graph

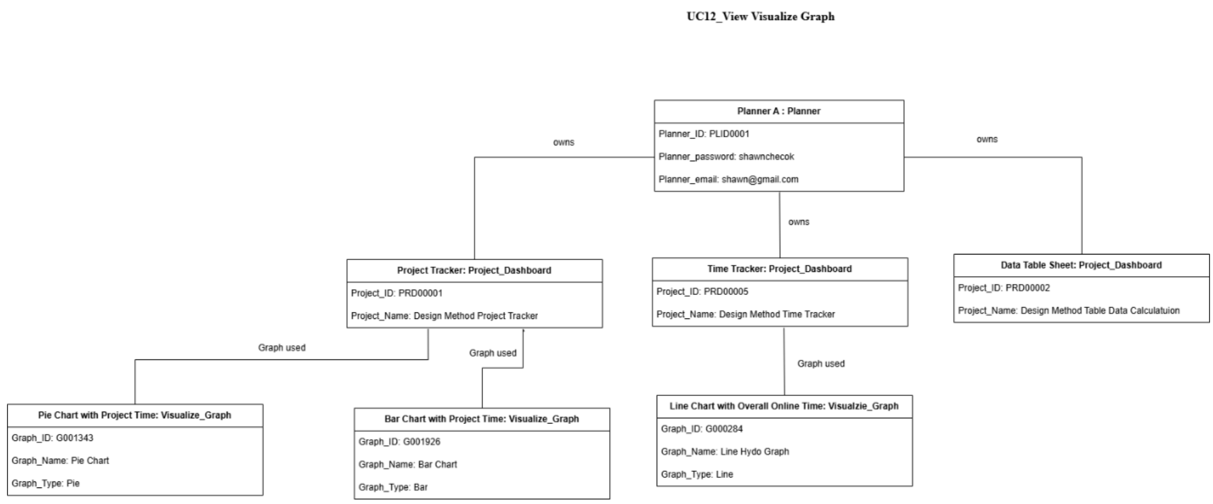

Figure 19 UC12_View Visualize Graph (Objective Diagram)

3.0 Conclusion

In conclusion, the system details of the Notion application were successfully to show out a few of the features and structures by using the design method. The functions in Notion have shown on creating the projects, Select Workspace, managing tasks, setting the times and dates, personalize notification, open relevant file with URL links, using AI to solve the questions, searching project dashboard, and view trash history. Apart from these functions, the improvement suggestion in the Notion extends two features are using real-time tracker and using visualize graph which have 12 features have to been design. The design UML diagram of the design method using are use case diagram, activity diagram, sequence diagram, state machine diagram, and object diagram.

In the use case diagram can visualize and document system functionality which can improve the communication and identifying system requirement. In activity diagram, the use case specification of each scenario have explained the details of postconditions, standard process, alternative flows, and exception flows. We also can view the overflow in the state of activity in this diagram to make sure the ideas of the system design are true. Besides, using the class diagram can measure the system development part can follow the object-oriented structure with using the class diagram. In the class diagram, I can learn the value, functions, methods, objects and their connection are suitable. In addition, the sequence diagram can learn the and visualize the interaction in the system which in the Object, message, states and time. Furthermore, the state machine diagram can provide a visual representation of states and transitions and make it easier to grasp the system such as the U11 using with time tracker. Lastly is the object diagram, it can provide a visual representation of specific instances of classes.

Although exception flows are described in the use cases, they are not clearly reflected in structural diagrams such as the activity diagram, or sequence diagrams. For the better system robustness and reliability, in the future improvements could include explicit modeling of error handling mechanisms. This could involve components like Error Logger, Validation Handler, or Retry Processor, which can manage failed API connections, invalid input, or data retrieval errors. Representing these modules visually will provide a more complete and fault-tolerant system architecture.

In this assignment, I have learn how to draw UML diagram and the system designing ideas and it methods. Through this process, I gained a deeper understanding of system design concepts and the methods used to model software systems effectively. I also learned how to analyze user requirements and translate them into structured design components. This assignment enhanced my critical thinking, especially in mapping user interactions and backend processes into visual diagrams. Overall, it has strengthened my ability to approach system design systematically and communicate technical ideas clearly through visual models.

4.0 References

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